

School of Computing
National University of Singapore
CS5340: Uncertainty Modeling in AI
Semester 1, AY 2025/26

Exercise 1 — Solutions

Question 1 (Lecture 2)

a)

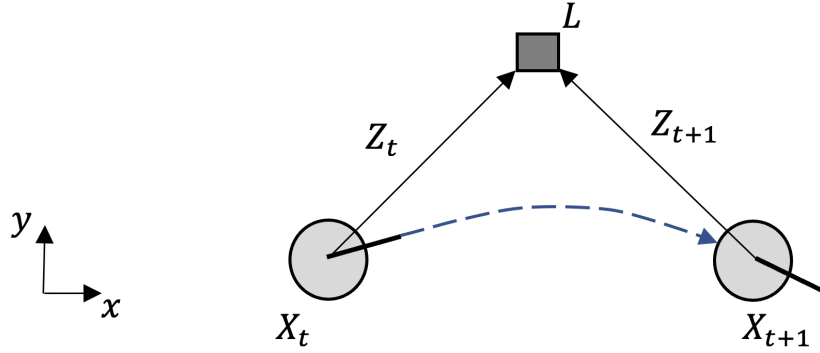


Figure 1.1

Figure 1.1 shows a mobile robot that traverses from pose X_t to X_{t+1} over time t to $t + 1$. The robot is equipped with an 1-dimensional range sensor that returns the distances Z_t and Z_{t+1} of a landmark structure L in the environment from the poses X_t and X_{t+1} respectively. Let U_t denote the control command given by the user to move the robot from X_t to X_{t+1} .

- (i) Taking $\{U_t, L, X_t, X_{t+1}, Z_t, Z_{t+1}\}$ as random variables, state whether each of these random variables is an observed or latent/hidden random variable. Explain your answers.

Answer:

- Observed: $\{U_t, Z_t, Z_{t+1}\}$. U_t is the command given by the user (so its value is known), and Z_t, Z_{t+1} are the readings returned by the range sensor.
- Latent/hidden: $\{X_t, X_{t+1}, L\}$. The robot poses and the landmark position are never measured directly; they can only be inferred from the controls and the range measurements.

- (ii) Given the following conditional independencies:

$$L \perp U_t \mid \emptyset, \quad X_t \perp L \mid U_t, \quad X_{t+1} \perp \{L, U_t\} \mid X_t, \\ Z_t \perp \{U_t, X_{t+1}\} \mid \{X_t, L\}, \quad Z_{t+1} \perp \{U_t, X_t, Z_t\} \mid \{L, X_{t+1}\}.$$

Write the factorized probability and draw the Bayesian network that represents the joint distribution

$$p(u_t, l, x_t, x_{t+1}, z_t, z_{t+1})$$

assuming the following topological ordering of the random variables:

$$\{U_t, L, X_t, X_{t+1}, Z_t, Z_{t+1}\}.$$

Show all your workings clearly.

Answer:

From the chain rule, following the topological ordering:

$$\begin{aligned} p(u_t, l, x_t, x_{t+1}, z_t, z_{t+1}) &= p(u_t) p(l \mid u_t) p(x_t \mid u_t, l) p(x_{t+1} \mid u_t, l, x_t) \\ &\quad \times p(z_t \mid u_t, l, x_t, x_{t+1}) p(z_{t+1} \mid u_t, l, x_t, x_{t+1}, z_t). \end{aligned}$$

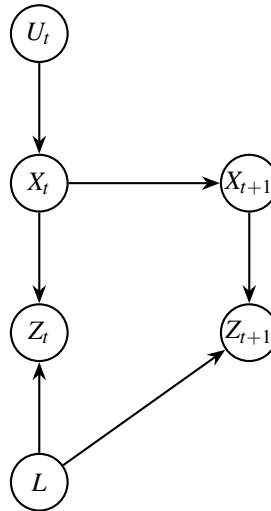
Each factor is simplified using the given conditional independencies:

$$\begin{aligned} p(l \mid u_t) &= p(l) && \text{since } L \perp U_t \mid \emptyset, \\ p(x_t \mid u_t, l) &= p(x_t \mid u_t) && \text{since } X_t \perp L \mid U_t, \\ p(x_{t+1} \mid u_t, l, x_t) &= p(x_{t+1} \mid x_t) && \text{since } X_{t+1} \perp \{L, U_t\} \mid X_t, \\ p(z_t \mid u_t, l, x_t, x_{t+1}) &= p(z_t \mid x_t, l) && \text{since } Z_t \perp \{U_t, X_{t+1}\} \mid \{X_t, L\}, \\ p(z_{t+1} \mid u_t, l, x_t, x_{t+1}, z_t) &= p(z_{t+1} \mid l, x_{t+1}) && \text{since } Z_{t+1} \perp \{U_t, X_t, Z_t\} \mid \{L, X_{t+1}\}. \end{aligned}$$

Hence

$$\boxed{p(u_t, l, x_t, x_{t+1}, z_t, z_{t+1}) = p(u_t) p(l) p(x_t \mid u_t) p(x_{t+1} \mid x_t) p(z_t \mid x_t, l) p(z_{t+1} \mid x_{t+1}, l).}$$

The corresponding Bayesian network (each node's parents are the conditioning variables of its factor):



- (iii) Write the following probability distribution $p(z_t, z_{t+1} \mid l)$ in terms of the factorized probability obtained in (ii). Simplify your answer.

Answer:

$$p(z_t, z_{t+1} | l) = \frac{p(z_t, z_{t+1}, l)}{p(l)},$$

where

$$\begin{aligned} p(z_t, z_{t+1}, l) &= \sum_{u_t} \sum_{x_t} \sum_{x_{t+1}} p(u_t) p(l) p(x_t | u_t) p(x_{t+1} | x_t) p(z_t | x_t, l) p(z_{t+1} | x_{t+1}, l) \\ &= p(l) \sum_{x_t} \sum_{x_{t+1}} p(x_{t+1} | x_t) p(z_t | x_t, l) p(z_{t+1} | x_{t+1}, l) \underbrace{\sum_{u_t} p(u_t) p(x_t | u_t)}_{=p(x_t)}. \end{aligned}$$

Dividing by $p(l)$:

$$\begin{aligned} p(z_t, z_{t+1} | l) &= \sum_{x_t} \sum_{x_{t+1}} \left[\sum_{u_t} p(u_t) p(x_t | u_t) \right] p(x_{t+1} | x_t) p(z_t | x_t, l) p(z_{t+1} | x_{t+1}, l) \\ &= \sum_{x_t} \sum_{x_{t+1}} p(x_t) p(x_{t+1} | x_t) p(z_t | x_t, l) p(z_{t+1} | x_{t+1}, l). \end{aligned}$$

b)

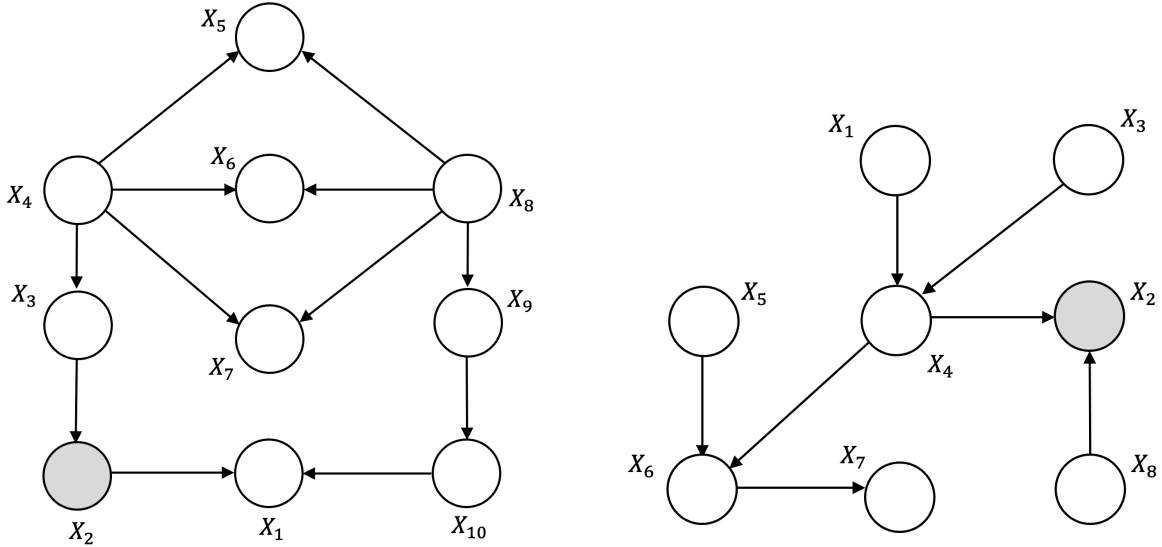


Figure 1.2

For each of the Bayesian networks shown in Figure 1.2, determine the largest set of nodes X_B such that $X_1 \perp X_B | X_2$. Explain your answers.

Answer:

(i) Left network: $X_B = \{X_3, X_4\}$, i.e. $X_1 \perp \{X_3, X_4\} | X_2$.

The path $X_1 \leftarrow X_2 \leftarrow X_3 \leftarrow X_4$ is blocked because X_2 is observed (head-to-tail at X_2). The path $X_1 \leftarrow X_{10} \leftarrow X_9 \leftarrow X_8$ is open, so the Bayes ball reaches X_{10}, X_9, X_8 , and from X_8 it reaches X_5, X_6, X_7 . However, it cannot pass from X_8 to X_4 , because X_5, X_6 and X_7 are all head-to-head nodes ($X_4 \rightarrow X_k \leftarrow X_8$) that are unobserved and have no observed descendants. Hence X_3 and X_4 are d-separated from X_1 given X_2 , and

all other nodes are not.

(ii) **Right network:** $X_B = \{X_5\}$, i.e. $X_1 \perp X_5 \mid X_2$.

The only path from X_1 to X_5 is $X_1 \rightarrow X_4 \rightarrow X_6 \leftarrow X_5$, which contains the head-to-head structure $X_4 \rightarrow X_6 \leftarrow X_5$. Since neither X_6 nor its descendant X_7 is observed, the path is blocked. All other nodes are d-connected to X_1 : X_4 , X_6 and X_7 via the unobserved chain $X_1 \rightarrow X_4 \rightarrow X_6 \rightarrow X_7$; X_3 via $X_1 \rightarrow X_4 \leftarrow X_3$, which is open because the observed X_2 is a descendant of the head-to-head node X_4 ; and X_8 via $X_1 \rightarrow X_4 \rightarrow X_2 \leftarrow X_8$, which is open because the head-to-head node X_2 is observed.

Question 2 (Lecture 4)

Consider the graph shown in Figure 2.1:

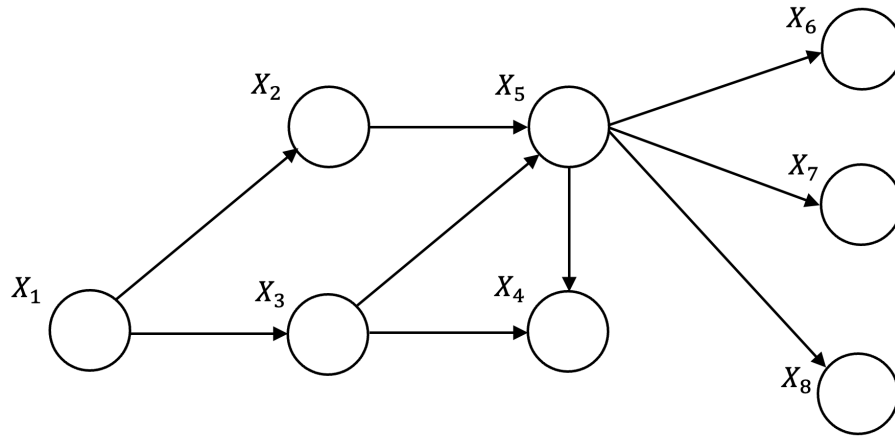
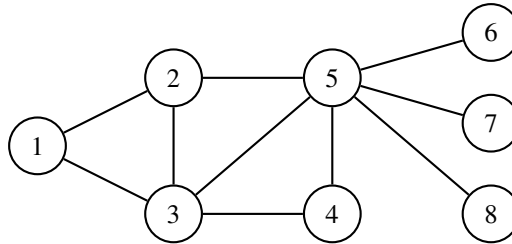


Figure 2.1

a) What is the corresponding moral graph?

Answer:

Drop the edge directions and “marry” the parents of every node. Node 5 has parents $\{2, 3\}$, so the edge $2 - 3$ is added; node 4 has parents $\{3, 5\}$, which are already connected.

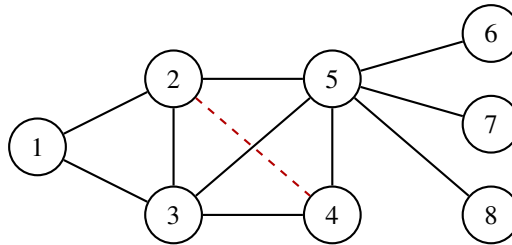


b) What is the reconstituted graph from the UNDIRECTEDGRAPHELMINATE algorithm on the moral graph with the ordering $\{8, 7, 6, 5, 4, 3, 2, 1\}$?

Answer:

Eliminating 8, 7 and 6 adds no edges (each has the single neighbour 5). Eliminating 5 (remaining neighbours $\{2, 3, 4\}$) adds the fill-in edge $2 - 4$ (dashed). Eliminating 4, 3, 2, 1 adds no further edges.

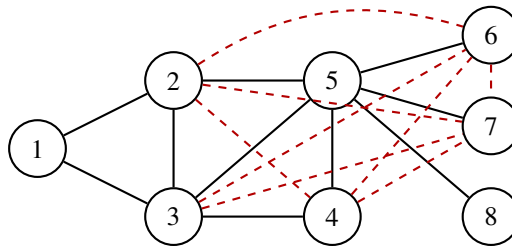
The elimination cliques are $\{5, 8\}, \{5, 7\}, \{5, 6\}, \{2, 3, 4, 5\}, \{2, 3, 4\}, \{1, 2, 3\}, \{1, 2\}, \{1\}$; the largest has size 4.



- c) What is the reconstituted graph from the `UNDIRECTEDGRAPHELIMINATE` algorithm on the moral graph with the ordering $\{8, 5, 6, 7, 4, 3, 2, 1\}$?

Answer:

Eliminating 8 adds no edge. Eliminating 5 (remaining neighbours $\{2, 3, 4, 6, 7\}$) makes these five nodes fully connected, adding the fill-in edges $2-4$, $2-6$, $2-7$, $3-6$, $3-7$, $4-6$, $4-7$, $6-7$ (dashed). No further edges are added afterwards. The largest elimination clique is $\{2, 3, 4, 5, 6, 7\}$, of size 6.



- d) Suppose you wish to calculate $p(x_1 \mid x_8)$. Which ordering is preferable? Why?

Answer:

The ordering in (b) is preferable. Its largest elimination clique has size 4, so the most expensive intermediate factor involves 4 variables (cost $O(k^4)$ for k -state variables). The ordering in (c) creates an elimination clique of size 6 (cost $O(k^6)$).

Question 3 (Lecture 4)

What is the treewidth of the graph shown in Figure 3.1?

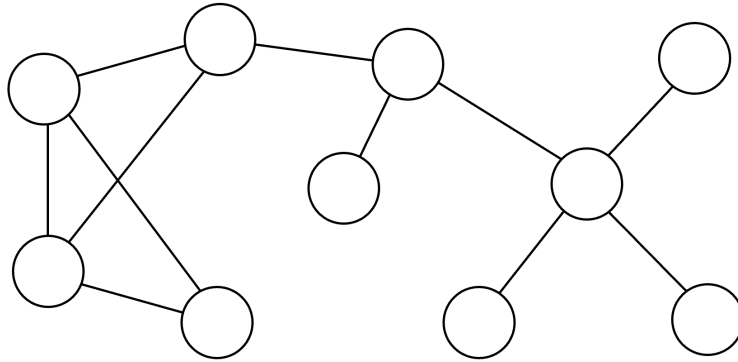
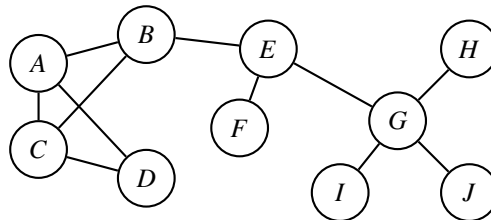


Figure 3.1

Answer:

- Eliminating a node adds edges between its (remaining) neighbours, which creates an “elimination clique” in the graph.
- Treewidth: one less than the smallest achievable cardinality of the largest elimination clique over all possible elimination orderings.

Label the nodes of Figure 3.1 as follows:



Use the elimination ordering $I = \{F, H, I, J, G, E, D, B, A, C\}$:

- F, H, I, J are leaves (a single neighbour): no fill-in edges, elimination cliques of size 2.
- G (remaining neighbour E) and then E (remaining neighbour B): no fill-in, cliques of size 2.
- D has neighbours $\{A, C\}$, which are already connected: clique $\{A, C, D\}$ of size 3, no fill-in.
- B has neighbours $\{A, C\}$, already connected: clique $\{A, B, C\}$ of size 3, no fill-in.
- A and C : cliques of size 2 and 1.

This ordering introduces no fill-in edges (the graph is already triangulated), and the largest elimination clique has size 3. No ordering can do better: the graph contains the triangle $A-B-C$, and whichever of A, B, C is eliminated first still has the other two as neighbours, so every ordering produces an elimination clique of size at least 3. Hence

$$\text{treewidth} = 3 - 1 = 2.$$

Question 4 (Lecture 2 and 3)

Consider the following random variables. X_1 and X_2 represent the outcomes of two independent fair coin tosses. X_3 is the indicator function of the event that the outcomes are identical.

- a) Specify a directed graphical model that describes the joint probability distribution (i.e. specify the graph and the conditional distributions).

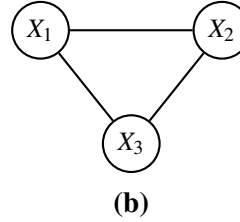
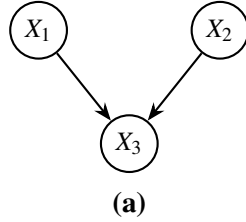
Answer:

Let 1 denote heads and 0 denote tails. The directed model is the v-structure $X_1 \rightarrow X_3 \leftarrow X_2$ (graph (a) below), with

$$P(X_1, X_2, X_3) = P(X_1)P(X_2)P(X_3 | X_1, X_2),$$

where $P(X_1 = 1) = P(X_1 = 0) = \frac{1}{2}$, $P(X_2 = 1) = P(X_2 = 0) = \frac{1}{2}$, and

X_1	X_2	$P(X_3 = 1 X_1, X_2)$	$P(X_3 = 0 X_1, X_2)$
0	0	1	0
0	1	0	1
1	0	0	1
1	1	1	0



- b) Specify an undirected graphical model that describes the joint probability distribution (i.e. give the graph and specify the clique potentials).

Answer:

See graph (b) above. The directed model in (a) has a factor involving all three variables, so the undirected model must have a clique containing X_1 , X_2 and X_3 (this is the moral graph of (a)). With a single clique:

$$P(X_1, X_2, X_3) = \frac{1}{Z} \Phi_1(X_1, X_2, X_3),$$

where $\Phi_1(X_1, X_2, X_3) = P(X_1)P(X_2)P(X_3 | X_1, X_2)$ and $Z = 1$. Explicitly, $\Phi_1 = \frac{1}{4}$ for the configurations $(0, 0, 1)$, $(0, 1, 0)$, $(1, 0, 0)$, $(1, 1, 1)$ and $\Phi_1 = 0$ otherwise.

- c) In both cases, list all conditional independencies that are implied by the graph.

Answer:

Directed graph (a): only $X_1 \perp X_2 | \emptyset$ (the unobserved head-to-head node X_3 blocks the path). Undirected graph (b): the graph is complete, so it implies *no* conditional independence statements.

- d) In both cases, list any additional conditional dependencies that are displayed by this probability distribution but are not implied by the graph.

Answer:

Besides $X_1 \perp X_2 \mid \emptyset$, the distribution also satisfies $X_1 \perp X_3 \mid \emptyset$ and $X_2 \perp X_3 \mid \emptyset$. To see this, note that $P(X_1=0) = P(X_1=1) = P(X_2=0) = P(X_2=1) = P(X_3=0) = P(X_3=1) = \frac{1}{2}$, and $P(x_1, x_3) = \frac{1}{4} = P(x_1)P(x_3)$ for all values of x_1, x_3 . Similarly $P(x_2, x_3) = \frac{1}{4} = P(x_2)P(x_3)$ for all values of x_2, x_3 .

This is a case where the three random variables are pairwise independent but not mutually independent (e.g. $P(x_1, x_2, x_3) \neq P(x_1)P(x_2)P(x_3)$). Neither graph implies $X_1 \perp X_3$ or $X_2 \perp X_3$ (and the undirected graph does not imply $X_1 \perp X_2$ either). These independencies are numerical rather than structural: a slight change in the parameters (e.g. slightly unfair coins) destroys them.

Question 5 (Lecture 2)

The graphical model shown in Figure 5.1 describes some relationships among variables associated with chest abnormality. Answer the following questions based on the graphical model.

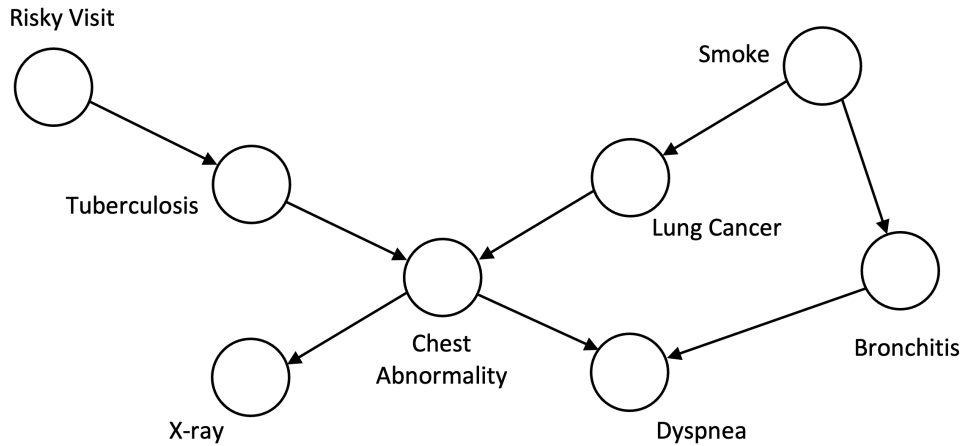


Figure 5.1

- a) True or False. Justify your choice. $Smoke \perp Dyspnea \mid Bronchitis$.

Answer:

False. *Bronchitis* blocks the path $Smoke \rightarrow Bronchitis \rightarrow Dyspnea$, but the path $Smoke \rightarrow Lung\ Cancer \rightarrow Chest\ Abnormality \rightarrow Dyspnea$ contains no observed node and no head-to-head node, so it remains open.

- b) True or False. Justify your choice. $Bronchitis \perp X-ray \mid Cancer$.

Answer:

True. The path $Bronchitis \leftarrow Smoke \rightarrow Lung\ Cancer \rightarrow Chest\ Abnormality \rightarrow X-ray$ is blocked by the observed *Cancer* (head-to-tail). The path $Bronchitis \rightarrow Dyspnea \leftarrow Chest\ Abnormality \rightarrow X-ray$ is blocked by the unobserved head-to-head node *Dyspnea*.

- c) True or False. Justify your choice. $Smoke \perp Risky\ Visit \mid Dyspnea$.

Answer:

False. *Dyspnea* is a descendant of the head-to-head node *Chest Abnormality*, so observing it opens the path *Smoke* $\rightarrow$ *Lung Cancer* $\rightarrow$ *Chest Abnormality* $\leftarrow$ *Tuberculosis* $\leftarrow$ *Risky Visit*. (The path through *Bronchitis* $\rightarrow$ *Dyspnea* $\leftarrow$ *Chest Abnormality* is also opened, since the head-to-head node *Dyspnea* is itself observed.)

d) True or False. Justify your choice. $X\text{-ray} \perp \text{Smoke} \mid \{\text{Cancer}, \text{Bronchitis}\}$.

Answer:

True. The path $X\text{-ray} \leftarrow \text{Chest Abnormality} \leftarrow \text{Lung Cancer} \leftarrow \text{Smoke}$ is blocked by the observed *Lung Cancer*. The path $X\text{-ray} \leftarrow \text{Chest Abnormality} \rightarrow \text{Dyspnea} \leftarrow \text{Bronchitis} \leftarrow \text{Smoke}$ is blocked by the observed *Bronchitis* (and also by the unobserved head-to-head node *Dyspnea*).

Question 6 (Lecture 4)

Evaluate (give the distribution tables) the following probabilities:

$$p(x_1 \mid x_5), p(x_2 \mid x_4), p(x_3 \mid x_2), p(x_4 \mid x_3), p(x_5)$$

for the Bayesian network shown in Figure 6.1, where each random variable takes a binary state, i.e. $x_i \in \{T, F\}$. Show all your workings clearly.

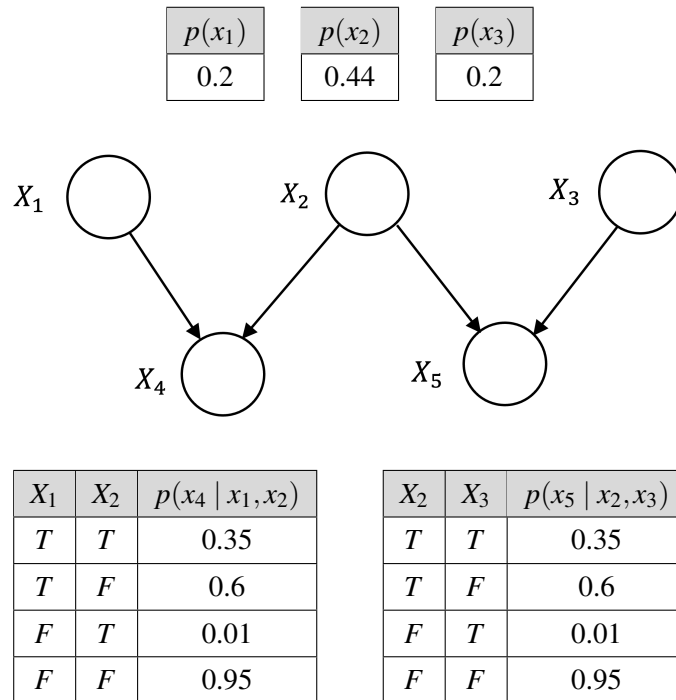


Figure 6.1

Answer:

Throughout, $1 \equiv T$ and $0 \equiv F$, and the tables give the probability of the state *T*, e.g. $p(x_1=1) = 0.2$,

$p(x_1=0) = 0.8, p(x_2=1) = 0.44, p(x_2=0) = 0.56, p(x_3=1) = 0.2, p(x_3=0) = 0.8$. The joint distribution is

$$p(x_1, \dots, x_5) = p(x_1) p(x_2) p(x_3) p(x_4 | x_1, x_2) p(x_5 | x_2, x_3).$$

For $p(x_2 | x_4)$:

$$p(x_2 | x_4) = \frac{p(x_2, x_4)}{\sum_{x_2} p(x_2, x_4)} = \frac{p(x_2, x_4)}{p(x_4)},$$

$$\begin{aligned} p(x_2, x_4) &= \sum_{x_1} \sum_{x_3} \sum_{x_5} p(x_1) p(x_2) p(x_3) p(x_4 | x_1, x_2) p(x_5 | x_2, x_3) = \sum_{x_1} p(x_1) p(x_2) p(x_4 | x_1, x_2) \\ &= p(x_2) \{ p(x_1=0) p(x_4 | x_1=0, x_2) + p(x_1=1) p(x_4 | x_1=1, x_2) \}, \end{aligned}$$

using $\sum_{x_3} \sum_{x_5} p(x_3) p(x_5 | x_2, x_3) = 1$.

x_1	x_2	$p(x_4=1 x_1, x_2)$	$p(x_4=0 x_1, x_2)$
<i>T</i>	<i>T</i>	0.35	0.65
<i>T</i>	<i>F</i>	0.6	0.4
<i>F</i>	<i>T</i>	0.01	0.99
<i>F</i>	<i>F</i>	0.95	0.05

$$\begin{aligned} p(x_2=0, x_4=0) &= (0.56) \{ (0.8)(0.05) + (0.2)(0.4) \} = 0.0672, \\ p(x_2=0, x_4=1) &= (0.56) \{ (0.8)(0.95) + (0.2)(0.6) \} = 0.4928, \\ p(x_2=1, x_4=0) &= (0.44) \{ (0.8)(0.99) + (0.2)(0.65) \} = 0.4057, \\ p(x_2=1, x_4=1) &= (0.44) \{ (0.8)(0.01) + (0.2)(0.35) \} = 0.0343. \end{aligned}$$

Since $p(x_4) = \sum_{x_2} p(x_2, x_4)$:

$$\begin{aligned} p(x_4=0) &= 0.0672 + 0.4057 = 0.4729, \\ p(x_4=1) &= 0.4928 + 0.0343 = 0.5271. \end{aligned}$$

Hence

x_2	x_4	$p(x_2, x_4)$	$p(x_2 x_4) = p(x_2, x_4) / p(x_4)$
0	0	0.0672	$0.0672 / 0.4729 = \mathbf{0.1421}$
0	1	0.4928	$0.4928 / 0.5271 = \mathbf{0.9349}$
1	0	0.4057	$0.4057 / 0.4729 = \mathbf{0.8579}$
1	1	0.0343	$0.0343 / 0.5271 = \mathbf{0.0651}$

For $p(x_1 | x_5), p(x_3 | x_2), p(x_4 | x_3)$:

- $p(x_1 | x_5) = p(x_1)$: the only path $X_1 \rightarrow X_4 \leftarrow X_2 \rightarrow X_5$ is blocked by the unobserved head-to-head node X_4 .
- $p(x_3 | x_2) = p(x_3)$: the only path $X_2 \rightarrow X_5 \leftarrow X_3$ is blocked by the unobserved head-to-head node X_5 .

- $p(x_4 | x_3) = p(x_4)$: the only path $X_4 \leftarrow X_2 \rightarrow X_5 \leftarrow X_3$ is blocked by the unobserved head-to-head node X_5 ($p(x_4)$ was computed above).

	$p(x_1 x_5) = p(x_1)$	$p(x_3 x_2) = p(x_3)$	$p(x_4 x_3) = p(x_4)$
$= 1 (T)$	0.2	0.2	0.5271
$= 0 (F)$	0.8	0.8	0.4729

(Each column holds for both values of the conditioning variable.)

For $p(x_5)$:

$$p(x_5) = \sum_{x_1} \sum_{x_2} \sum_{x_3} \sum_{x_4} p(x_1) p(x_2) p(x_3) p(x_4 | x_1, x_2) p(x_5 | x_2, x_3) = \sum_{x_2} \sum_{x_3} p(x_2) p(x_3) p(x_5 | x_2, x_3).$$

x_2	x_3	$p(x_5=1 x_2, x_3)$	$p(x_5=0 x_2, x_3)$
T	T	0.35	0.65
T	F	0.6	0.4
F	T	0.01	0.99
F	F	0.95	0.05

x_2	x_3	$p(x_2)p(x_3)p(x_5=0 x_2, x_3)$	$p(x_2)p(x_3)p(x_5=1 x_2, x_3)$
0	0	$(0.56)(0.8)(0.05) = 0.0224$	$(0.56)(0.8)(0.95) = 0.4256$
0	1	$(0.56)(0.2)(0.99) = 0.11088$	$(0.56)(0.2)(0.01) = 0.00112$
1	0	$(0.44)(0.8)(0.40) = 0.1408$	$(0.44)(0.8)(0.60) = 0.2112$
1	1	$(0.44)(0.2)(0.65) = 0.0572$	$(0.44)(0.2)(0.35) = 0.0308$

x_5	$p(x_5)$
0	$0.0224 + 0.11088 + 0.1408 + 0.0572 = \mathbf{0.33128}$
1	$0.4256 + 0.00112 + 0.2112 + 0.0308 = \mathbf{0.66872}$

Question 7 (Lecture 5)

Give the junction tree of the Bayesian network shown in Figure 7.1 using the following elimination order: $\{X_7, X_6, X_5, X_4, X_3, X_2, X_1\}$. Show all your workings clearly.

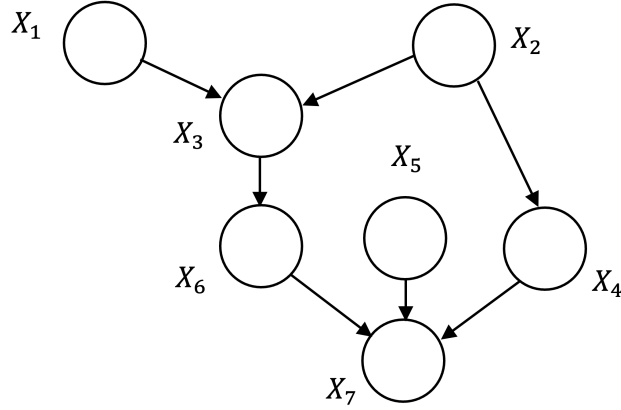


Figure 7.1

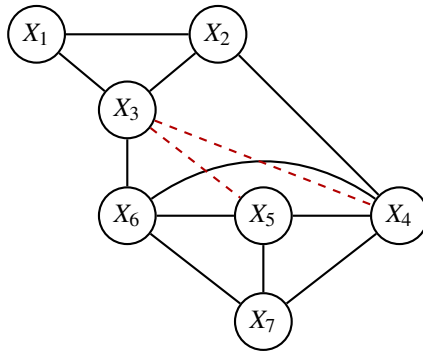
Answer:

1. Moralization: drop all edge directions and marry the parents of every node: $X_1 - X_2$ (parents of X_3), and $X_4 - X_5$, $X_4 - X_6$, $X_5 - X_6$ (parents of X_7).

2. Triangulation with the elimination order $\{X_7, X_6, X_5, X_4, X_3, X_2, X_1\}$:

- Eliminate X_7 : neighbours $\{X_4, X_5, X_6\}$ are already fully connected, so there is no fill-in.
- Eliminate X_6 : neighbours $\{X_3, X_4, X_5\}$; add the fill-in edges $X_3 - X_4$ and $X_3 - X_5$.
- Eliminate X_5 : neighbours $\{X_3, X_4\}$; no fill-in.
- Eliminate X_4 : neighbours $\{X_2, X_3\}$; no fill-in.
- Eliminate X_3 : neighbours $\{X_1, X_2\}$; no fill-in. Then X_2 and X_1 .

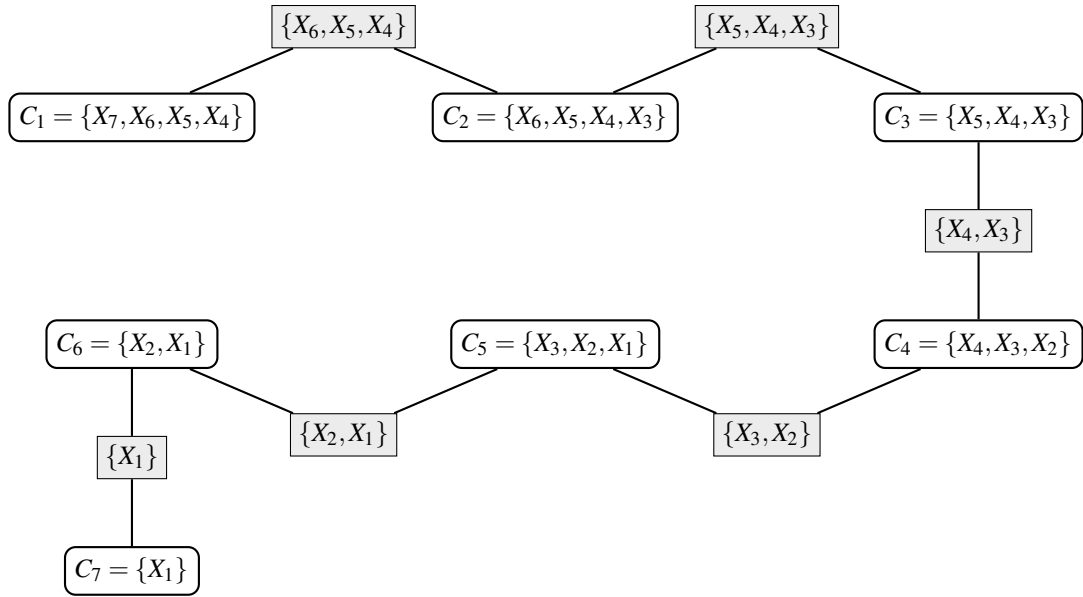
Triangulated graph (moral edges solid, fill-in edges dashed):



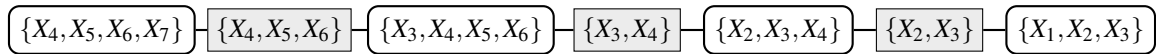
3. Form clusters from the elimination cliques:

$$C_1 : \{X_7, X_6, X_5, X_4\}, \quad C_2 : \{X_6, X_5, X_4, X_3\}, \quad C_3 : \{X_5, X_4, X_3\}, \quad C_4 : \{X_4, X_3, X_2\}, \\ C_5 : \{X_3, X_2, X_1\}, \quad C_6 : \{X_2, X_1\}, \quad C_7 : \{X_1\}.$$

4. Get the maximum spanning tree of the cluster graph (edge weight = size of the sepset). This gives the junction tree:



Note: C_3 , C_6 and C_7 are subsets of neighbouring clusters and can be absorbed. This gives the equivalent junction tree over the maximal cliques only:



Question 8 (Lecture 2 and 5)

Figure 8.1 shows a Bayesian network with five random variables X_1, X_2, X_3, X_4, X_5 , where $x_i \in \{0, 1\}$ for $i = 1, 2, 4$, and $x_i \in \{0, 1, 2\}$ for $i = 3, 5$.

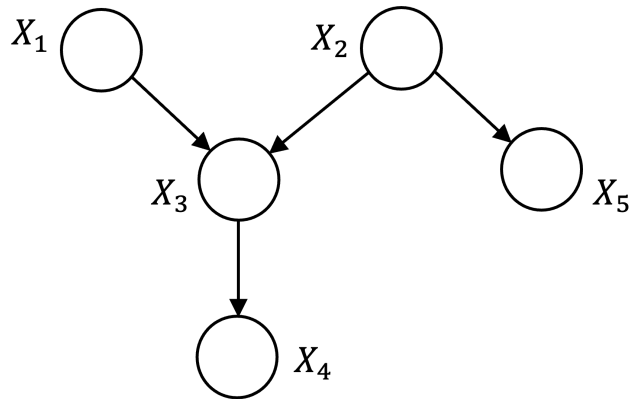


Figure 8.1

- Write down all the conditional independences given by the Bayesian network.
- Write down the factorized expression of the joint probability given by the Bayesian network.
- Convert the Bayesian network into a factor graph. Draw the factor graph and write down the expression of each factor clearly in your answer.

- (d) Table 8.1 gives the probability tables of the Bayesian network, find the conditional probability $p(x_1 \mid x_3 = 1, x_2)$. Show all your workings clearly.

X_1	X_2	X_3	$p(x_3 \mid x_1, x_2)$
0	0	0	0.3
0	0	1	0.4
0	1	0	0.9
0	1	1	0.08
1	0	0	0.05
1	0	1	0.25
1	1	0	0.5
1	1	1	0.3

X_1	$p(x_1)$	X_2	$p(x_2)$
0	0.6	0	0.7

X_3	X_4	$p(x_4 \mid x_3)$
0	0	0.1
1	0	0.4
2	0	0.99

Table 8.1

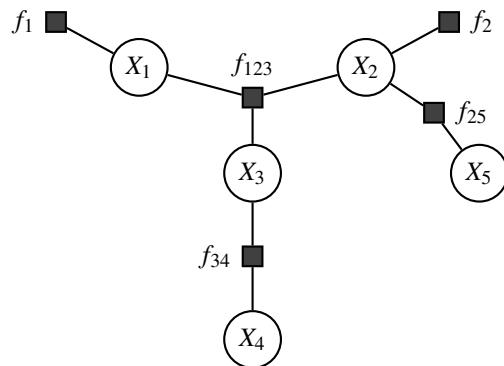
Answer:

- (a) The conditional independences (each node is independent of its non-descendants given its parents):

$$\begin{aligned}
 X_1 &\perp X_2 \mid \emptyset, & X_1 &\perp \{X_2, X_5\} \mid \emptyset, \\
 X_3 &\perp X_5 \mid \{X_1, X_2\}, \\
 X_4 &\perp \{X_1, X_2, X_5\} \mid X_3, \\
 X_5 &\perp \{X_1, X_3, X_4\} \mid X_2.
 \end{aligned}$$

(b) $p(X) = p(x_1) p(x_2) p(x_3 \mid x_1, x_2) p(x_4 \mid x_3) p(x_5 \mid x_2)$.

- (c) Factor graph and factors:



Factors:

$$\begin{aligned}
 f_1 &= p(x_1) \\
 f_2 &= p(x_2) \\
 f_{123} &= p(x_3 \mid x_1, x_2) \\
 f_{34} &= p(x_4 \mid x_3) \\
 f_{25} &= p(x_5 \mid x_2)
 \end{aligned}$$

- (d)

$$p(x_1 \mid x_3 = 1, x_2) = \frac{p(x_1, x_2, x_3 = 1)}{p(x_2, x_3 = 1)},$$

$$\begin{aligned}
p(x_1, x_2, x_3 = 1) &= \sum_{x_4} \sum_{x_5} p(x_1) p(x_2) p(x_3 = 1 \mid x_1, x_2) p(x_4 \mid x_3 = 1) p(x_5 \mid x_2) \\
&= p(x_1) p(x_2) p(x_3 = 1 \mid x_1, x_2) \underbrace{\sum_{x_4} p(x_4 \mid x_3 = 1)}_{=1} \underbrace{\sum_{x_5} p(x_5 \mid x_2)}_{=1} \\
&= p(x_1) p(x_2) p(x_3 = 1 \mid x_1, x_2).
\end{aligned}$$

X_1	X_2	$p(x_1, x_2, x_3=1) = p(x_1) p(x_2) p(x_3=1 \mid x_1, x_2)$
0	0	$(0.6)(0.7)(0.4) = \mathbf{0.168}$
0	1	$(0.6)(0.3)(0.08) = \mathbf{0.0144}$
1	0	$(0.4)(0.7)(0.25) = \mathbf{0.07}$
1	1	$(0.4)(0.3)(0.3) = \mathbf{0.036}$

$$p(x_2, x_3 = 1) = \sum_{x_1} p(x_1, x_2, x_3 = 1) = p(x_2) \sum_{x_1} p(x_1) p(x_3 = 1 \mid x_1, x_2).$$

X_2	$p(x_2, x_3=1)$
0	$p(x_2=0)\{p(x_1=0) p(x_3=1 \mid x_1=0, x_2=0) + p(x_1=1) p(x_3=1 \mid x_1=1, x_2=0)\}$ $= (0.7)\{(0.6)(0.4) + (0.4)(0.25)\} = \mathbf{0.238}$
1	$p(x_2=1)\{p(x_1=0) p(x_3=1 \mid x_1=0, x_2=1) + p(x_1=1) p(x_3=1 \mid x_1=1, x_2=1)\}$ $= (0.3)\{(0.6)(0.08) + (0.4)(0.3)\} = \mathbf{0.0504}$

X_1	X_2	$p(x_1 \mid x_3=1, x_2) = \frac{p(x_1, x_2, x_3=1)}{p(x_2, x_3=1)}$
0	0	$0.168/0.238 = \mathbf{0.706}$
0	1	$0.0144/0.0504 = \mathbf{0.286}$
1	0	$0.07/0.238 = \mathbf{0.294}$
1	1	$0.036/0.0504 = \mathbf{0.714}$

Question 9 (Lecture 5)

Figure 9.1 shows a graphical model with six binary-state latent random variables $Z = \{Z_1, Z_2, Z_3, Z_4, Z_5, Z_6\}$, $z_i \in \{0, 1\}$, and six binary-state observed random variables $X = \{X_1, X_2, X_3, X_4, X_5, X_6\}$, $x_i \in \{0, 1\}$. Table 9.1 gives the pairwise potentials $\phi(z_i, z_j)$, $\forall i, j \in \mathcal{E}_Z$ and conditional probability $p(x_i \mid z_i)$ for $i = 1, \dots, 6$, where $\mathcal{E}_Z$ denotes all the edges between the latent random variables in the graphical model. Find the configuration of Z that maximizes the joint probability $p(X, Z)$.

Hint: convert the graphical model into a factor graph, where the respective pairwise potential and conditional probability are represented as a single factor.

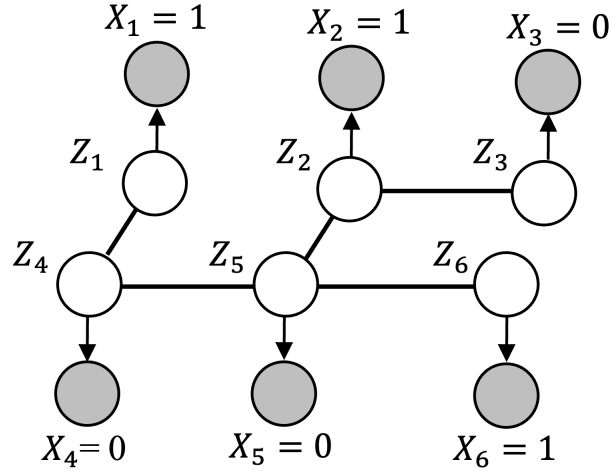


Figure 9.1

Z_i	Z_j	$\phi(z_i, z_j)$	X_i	Z_i	$p(x_i z_i)$
0	0	0	0	0	0.9
0	1	2	0	1	0.05
1	0	2	1	0	0.1
1	1	0	1	1	0.95

Table 9.1

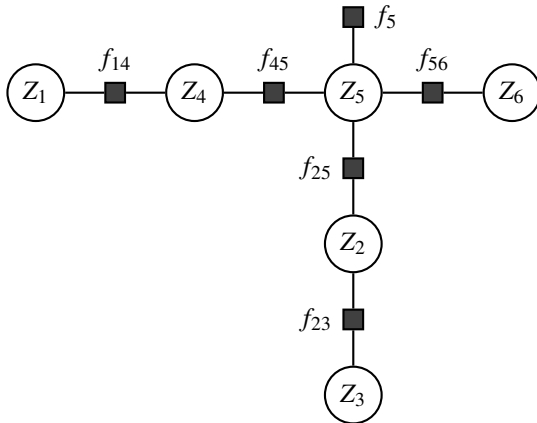
Answer:

Joint probability:

$$p(X, Z) = \frac{1}{Z_p} \prod_{i=1}^6 p(x_i | z_i) \phi(z_1, z_4) \phi(z_4, z_5) \phi(z_2, z_5) \phi(z_2, z_3) \phi(z_5, z_6),$$

with the observations $x_1 = 1, x_2 = 1, x_3 = 0, x_4 = 0, x_5 = 0, x_6 = 1$.

Convert the graphical model into a factor graph:



Factors:

$$\begin{aligned} f_{14} &= p(x_1=1 | z_1) \phi(z_1, z_4) \\ f_{45} &= p(x_4=0 | z_4) \phi(z_4, z_5) \\ f_{23} &= p(x_3=0 | z_3) \phi(z_2, z_3) \\ f_{25} &= p(x_2=1 | z_2) \phi(z_2, z_5) \\ f_{56} &= p(x_6=1 | z_6) \phi(z_5, z_6) \\ f_5 &= p(x_5=0 | z_5) \end{aligned}$$

Apply the max-product algorithm on the factor graph with Z_5 as the root.

$$m_{z_1}^{\max}(z_4) = \max_{z_1} \mu_{f_{14} \rightarrow z_4} = \max_{z_1} p(x_1=1 \mid z_1) \phi(z_1, z_4)$$

Z_1	Z_4	$p(x_1=1 \mid z_1) \phi(z_1, z_4)$
0	0	$0.1 \times 0 = 0$
0	1	$0.1 \times 2 = 0.2$
1	0	$0.95 \times 2 = 1.9$
1	1	$0.95 \times 0 = 0$

$\delta^{\max}(z_1)$	Z_4	$m_{z_1}^{\max}(z_4)$
1	0	1.9
0	1	0.2

$$m_{z_6}^{\max}(z_5) = \max_{z_6} \mu_{f_{56} \rightarrow z_5} = \max_{z_6} p(x_6=1 \mid z_6) \phi(z_5, z_6)$$

Z_5	Z_6	$p(x_6=1 \mid z_6) \phi(z_5, z_6)$
0	0	$0.1 \times 0 = 0$
0	1	$0.95 \times 2 = 1.9$
1	0	$0.1 \times 2 = 0.2$
1	1	$0.95 \times 0 = 0$

$\delta^{\max}(z_6)$	Z_5	$m_{z_6}^{\max}(z_5)$
1	0	1.9
0	1	0.2

$$m_{z_3}^{\max}(z_2) = \max_{z_3} \mu_{f_{23} \rightarrow z_2} = \max_{z_3} p(x_3=0 \mid z_3) \phi(z_2, z_3)$$

Z_2	Z_3	$p(x_3=0 \mid z_3) \phi(z_2, z_3)$
0	0	$0.9 \times 0 = 0$
0	1	$0.05 \times 2 = 0.1$
1	0	$0.9 \times 2 = 1.8$
1	1	$0.05 \times 0 = 0$

$\delta^{\max}(z_3)$	Z_2	$m_{z_3}^{\max}(z_2)$
1	0	0.1
0	1	1.8

$$m_{z_2}^{\max}(z_5) = \max_{z_2} \mu_{f_{25} \rightarrow z_5} = \max_{z_2} p(x_2=1 \mid z_2) \phi(z_2, z_5) m_{z_3}^{\max}(z_2)$$

Z_2	Z_5	$p(x_2=1 \mid z_2) \phi(z_2, z_5) m_{z_3}^{\max}(z_2)$
0	0	$0.1 \times 0 \times 0.1 = 0$
0	1	$0.1 \times 2 \times 0.1 = 0.02$
1	0	$0.95 \times 2 \times 1.8 = 3.42$
1	1	$0.95 \times 0 \times 1.8 = 0$

$\delta^{\max}(z_2)$	Z_5	$m_{z_2}^{\max}(z_5)$
1	0	3.42
0	1	0.02

$$m_{z_4}^{\max}(z_5) = \max_{z_4} \mu_{f_{45} \rightarrow z_5} = \max_{z_4} p(x_4=0 \mid z_4) \phi(z_4, z_5) m_{z_1}^{\max}(z_4)$$

Z_4	Z_5	$p(x_4=0 \mid z_4) \phi(z_4, z_5) m_{z_1}^{\max}(z_4)$
0	0	$0.9 \times 0 \times 1.9 = 0$
0	1	$0.9 \times 2 \times 1.9 = 3.42$
1	0	$0.05 \times 2 \times 0.2 = 0.02$
1	1	$0.05 \times 0 \times 0.2 = 0$

$\delta^{\max}(z_4)$	Z_5	$m_{z_4}^{\max}(z_5)$
1	0	0.02
0	1	3.42

At the root:

$$\max_{z_5} m_{z_4}^{\max}(z_5) m_{z_2}^{\max}(z_5) m_{z_6}^{\max}(z_5) p(x_5=0 \mid z_5)$$

Z_5	$m_{z_4}^{\max}(z_5) m_{z_2}^{\max}(z_5) m_{z_6}^{\max}(z_5) p(x_5=0 \mid z_5)$
0	$0.02 \times 3.42 \times 1.9 \times 0.9 = \mathbf{0.117}$
1	$3.42 \times 0.02 \times 0.2 \times 0.05 = 0.000684$

Maximal (unnormalized) probability = 0.117, i.e. $\max_Z p(X, Z) = 0.117/Z_p$, attained at $z_5^{\max} = 0$.

Backtrack to get all the states that give the maximal probability:

$$\begin{aligned} z_5^{\max} = 0 &\Rightarrow z_4^{\max} = \delta^{\max}(z_4) \big|_{z_5=0} = 1, \quad z_2^{\max} = \delta^{\max}(z_2) \big|_{z_5=0} = 1, \quad z_6^{\max} = \delta^{\max}(z_6) \big|_{z_5=0} = 1, \\ z_4^{\max} = 1 &\Rightarrow z_1^{\max} = \delta^{\max}(z_1) \big|_{z_4=1} = 0, \\ z_2^{\max} = 1 &\Rightarrow z_3^{\max} = \delta^{\max}(z_3) \big|_{z_2=1} = 0. \end{aligned}$$

$$Z^{\max} = (z_1, z_2, z_3, z_4, z_5, z_6) = (0, 1, 0, 1, 0, 1)$$

Question 10 (Lecture 4 and 5)

Figure 10.1 shows a Bayesian Network with four random variables X_1, X_2, X_3 and X_4 , where $x_i \in \{0, 1\}$. The respective prior and conditional probability distribution tables are also given.

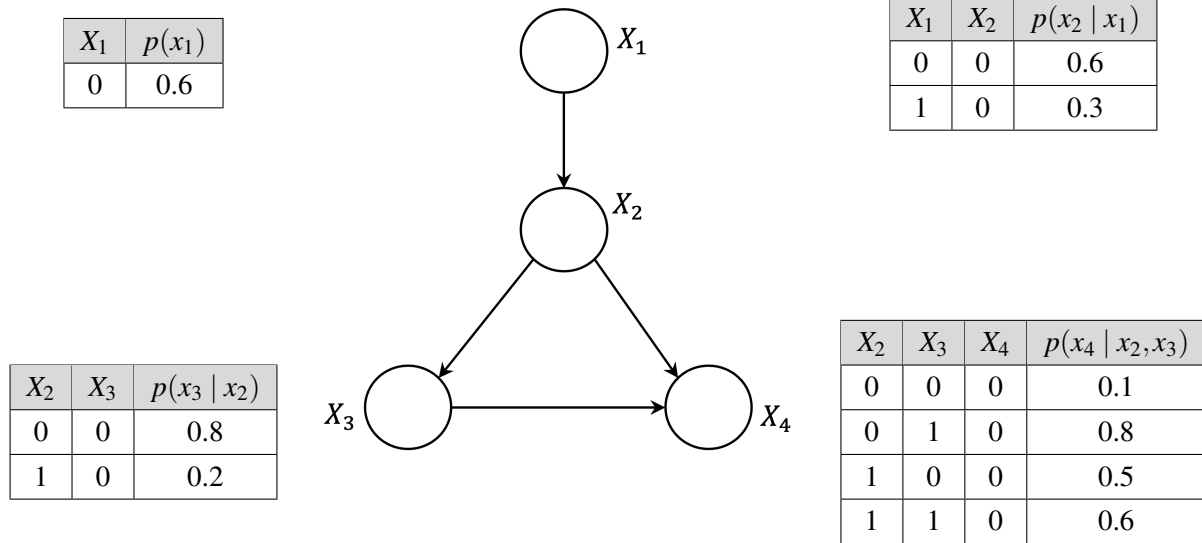


Figure 10.1

Find the following marginal probabilities:

- $p(x_2)$
- $p(x_3)$
- $p(x_4)$

d. $p(x_3, x_4)$

Answer:

Let $a = f(x_1, x_2) = p(x_1) p(x_2 | x_1)$ and $b = f(x_2, x_3, x_4) = p(x_3 | x_2) p(x_4 | x_2, x_3)$. The complete tables are:

x_1	$p(x_1)$
0	0.6
1	0.4

x_1	x_2	$p(x_2 x_1)$
0	0	0.6
0	1	0.4
1	0	0.3
1	1	0.7

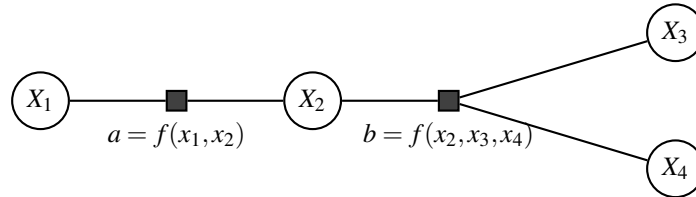
x_2	x_3	$p(x_3 x_2)$
0	0	0.8
0	1	0.2
1	0	0.2
1	1	0.8

x_2	x_3	x_4	$p(x_4 x_2, x_3)$
0	0	0	0.1
0	0	1	0.9
0	1	0	0.8
0	1	1	0.2
1	0	0	0.5
1	0	1	0.5
1	1	0	0.6
1	1	1	0.4

x_1	x_2	$f(x_1, x_2)$
0	0	0.36
0	1	0.24
1	0	0.12
1	1	0.28

x_2	x_3	x_4	$f(x_2, x_3, x_4)$
0	0	0	0.08
0	0	1	0.72
0	1	0	0.16
0	1	1	0.04
1	0	0	0.1
1	0	1	0.1
1	1	0	0.48
1	1	1	0.32

Factor graph:



Choose X_2 as the root node.

Upward pass

$$v_{1 \rightarrow a} = 1, \quad v_{3 \rightarrow b} = 1, \quad v_{4 \rightarrow b} = 1,$$

$$u_{a \rightarrow 2}(x_2) = \sum_{x_1} f(x_1, x_2) = f(x_1=0, x_2) + f(x_1=1, x_2) : \quad \begin{aligned} x_2 = 0 : 0.36 + 0.12 &= 0.48, \\ x_2 = 1 : 0.24 + 0.28 &= 0.52, \end{aligned}$$

$$u_{b \rightarrow 2}(x_2) = \sum_{x_3} \sum_{x_4} f(x_2, x_3, x_4) : \quad \begin{aligned} x_2 = 0 : 0.08 + 0.72 + 0.16 + 0.04 &= 1.0, \\ x_2 = 1 : 0.1 + 0.1 + 0.48 + 0.32 &= 1.0. \end{aligned}$$

a. $p(x_2) = u_{a \rightarrow 2}(x_2) \times u_{b \rightarrow 2}(x_2):$

x_2	$p(x_2)$
0	0.48
1	0.52

Downward pass

$$v_{2 \rightarrow a} = u_{b \rightarrow 2}(x_2), \quad v_{2 \rightarrow b} = u_{a \rightarrow 2}(x_2),$$

$$u_{a \rightarrow 1}(x_1) = \sum_{x_2} v_{2 \rightarrow a} f(x_1, x_2) = \sum_{x_2} 1.0 \times f(x_1, x_2) : \quad \begin{array}{l} x_1 = 0 : 0.36 + 0.24 = 0.6, \\ x_1 = 1 : 0.12 + 0.28 = 0.4, \end{array}$$

which recovers $p(x_1)$, as expected.

$$u_{b \rightarrow 3}(x_3) = \sum_{x_2} \sum_{x_4} u_{a \rightarrow 2}(x_2) f(x_2, x_3, x_4) :$$

$$x_3 = 0 : (0.48)(0.08) + (0.48)(0.72) + (0.52)(0.1) + (0.52)(0.1) = 0.488,$$

$$x_3 = 1 : (0.48)(0.16) + (0.48)(0.04) + (0.52)(0.48) + (0.52)(0.32) = 0.512.$$

b. $p(x_3) = u_{b \rightarrow 3}(x_3)$:

x_3	$p(x_3)$
0	0.488
1	0.512

$$u_{b \rightarrow 4}(x_4) = \sum_{x_2} \sum_{x_3} u_{a \rightarrow 2}(x_2) f(x_2, x_3, x_4) :$$

$$x_4 = 0 : (0.48)(0.08) + (0.48)(0.16) + (0.52)(0.1) + (0.52)(0.48) = 0.4168,$$

$$x_4 = 1 : (0.48)(0.72) + (0.48)(0.04) + (0.52)(0.1) + (0.52)(0.32) = 0.5832.$$

c. $p(x_4) = u_{b \rightarrow 4}(x_4)$:

x_4	$p(x_4)$
0	0.4168
1	0.5832

d.

$$\begin{aligned} p(x_3, x_4) &= \sum_{x_2} v_{2 \rightarrow b} f(x_2, x_3, x_4) = \sum_{x_2} u_{a \rightarrow 2}(x_2) f(x_2, x_3, x_4) \\ &= u_{a \rightarrow 2}(x_2=0) f(x_2=0, x_3, x_4) + u_{a \rightarrow 2}(x_2=1) f(x_2=1, x_3, x_4) : \end{aligned}$$

$$\begin{aligned}
(x_3, x_4) = (0, 0) &: (0.48)(0.08) + (0.52)(0.1) = 0.0904, \\
(x_3, x_4) = (0, 1) &: (0.48)(0.72) + (0.52)(0.1) = 0.3976, \\
(x_3, x_4) = (1, 0) &: (0.48)(0.16) + (0.52)(0.48) = 0.3264, \\
(x_3, x_4) = (1, 1) &: (0.48)(0.04) + (0.52)(0.32) = 0.1856.
\end{aligned}$$

x_3	x_4	$p(x_3, x_4)$
0	0	0.0904
0	1	0.3976
1	0	0.3264
1	1	0.1856

(Check: summing over x_4 gives $p(x_3) = (0.488, 0.512)$, and summing over x_3 gives $p(x_4) = (0.4168, 0.5832)$.)

Question 11 (Lecture 1, 2 and 3)

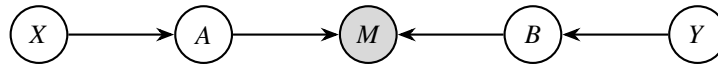
- a) Assume that the day of the week that females are born on, X , is independent of the day of the week, Y , on which males are born. Assume, however, that the old rhyme is true, and that personality is dependent on the day of the week you're born on. If A represents the female personality type and B the male personality type, then $(A \top X)$ and $(B \top Y)$, but $(A \perp B)$. Whether or not a male and a female are married, M , depends strongly on their personality types, $(M \top A, B)$, but is independent of X and Y if we know A and B . Draw a graphical model that can represent this setting. What can we say about the (graphical) dependency between the days of the week that John and Jane are born on, given that they are not married?

Show all your workings clearly.

Note: $(A \top X)$ is the shorthand for A is dependent on X .

Answer:

We have $A \top X$; $B \top Y$; $A \perp B$; $M \top \{A, B\}$; and $M \perp \{X, Y\} \mid \{A, B\}$. These are represented by the Bayesian network



with $p(X, Y, A, B, M) = p(X) p(Y) p(A \mid X) p(B \mid Y) p(M \mid A, B)$.

When M is unobserved, the only path $X \rightarrow A \rightarrow M \leftarrow B \leftarrow Y$ is blocked at the head-to-head node M , so $X \perp Y$ and $A \perp B$, as required.

When M is observed (John and Jane are *not* married), $A \rightarrow M \leftarrow B$ is a v-structure with an observed head-to-head node, so the path is no longer blocked: $A \not\perp B \mid M$. Intuitively, knowing they are not married tells us their personality types are likely to be incompatible. Since $A \top X$ and $B \top Y$ (the chains $X \rightarrow A$ and $B \leftarrow Y$ are unobserved), the whole path $X \rightarrow A \rightarrow M \leftarrow B \leftarrow Y$ is open, i.e.

$$X \not\perp Y \mid M.$$

The days of the week on which John and Jane were born become (graphically) **dependent** once we

know that they are not married, even though they are marginally independent.

b) Prove that the following relations are true or false:

(i) $(X \perp Y) \& (Y \perp Z) \implies (X \perp Z)$

(ii) $(X \perp Y \mid Z) \implies (X \perp Y, W \mid Z)$

(iii) Given that $(A \perp C \mid D, B) \& (A \perp B \mid D) \implies (A \perp B, C \mid D)$, does

$$(A \perp B \mid D, C) \& (A \perp C \mid D, B) \implies (A \perp B, C \mid D)$$

(iv) $(X, Y, Z \perp A, B, C \mid D, E, F) \implies (X \perp A, B \mid D, E, F) \& (X, Y \perp A \mid D, E, F) \dots$, i.e. (any subset of $\{X, Y, Z\} \perp$ any subset of $\{A, B, C\} \mid D, E, F$)

Show all your workings clearly.

Answer:

(i) **False.** Counter-example: let X and Y be two independent fair coins and let $Z = X$ (graphically: $X \rightarrow Z$, with Y disconnected). Then $X \perp Y$ and $Y \perp Z$, but $X \not\perp Z$, since $p(X=1, Z=1) = \frac{1}{2} \neq p(X=1)p(Z=1) = \frac{1}{4}$. (Many other counter-examples exist: any Z that depends on X but is independent of Y will do.)

(ii) **False.** Counter-example: let X, Y, Z be mutually independent fair coins and let $W = X$ (graphically: $X \rightarrow W$, with Y and Z disconnected). Then $X \perp Y \mid Z$, but $X \not\perp \{Y, W\} \mid Z$, since W carries full information about X : $p(X=1 \mid Y, W=1, Z) = 1 \neq p(X=1 \mid Z) = \frac{1}{2}$.

(iii) **True** (for strictly positive distributions; this is the *Intersection* property). Given $(A \perp B \mid D, C)$ and $(A \perp C \mid D, B)$, we have

$$p(A \mid D, C) = p(A \mid B, D, C) = p(A \mid B, D).$$

Applying the product rule to the first and last terms:

$$\begin{aligned} \frac{p(A, C \mid D)}{p(C \mid D)} &= \frac{p(A, B \mid D)}{p(B \mid D)} \implies p(B \mid D) p(A, C \mid D) = p(C \mid D) p(A, B \mid D) \\ &\implies \sum_C p(B \mid D) p(A, C \mid D) = \sum_C p(C \mid D) p(A, B \mid D) \\ &\implies p(B \mid D) p(A \mid D) = p(A, B \mid D), \quad \text{i.e. } (A \perp B \mid D). \end{aligned}$$

Hence $(A \perp B \mid D)$ and $(A \perp C \mid D, B)$, which by the given result (Contraction) yields $(A \perp B, C \mid D)$.

Remark: the equality $p(A \mid D, C) = p(A \mid B, D)$ must hold for *every* combination of (b, c, d) before we can sum over C . This is guaranteed when $p > 0$. Without positivity the result can fail: e.g. $A = B = C$ (one fair coin copied three times) with $D = \emptyset$ satisfies both premises, but $A \not\perp \{B, C\}$.

(iv) **True.** Let $\mathbb{M} = \{X, Y, Z\}$, $\mathbb{N} = \{A, B, C\}$ and $\mathbb{P} = \{D, E, F\}$. The statement is $\mathbb{M} \perp \mathbb{N} \mid \mathbb{P}$, i.e.

$$p(X, Y, Z, A, B, C \mid D, E, F) = p(X, Y, Z \mid D, E, F) p(A, B, C \mid D, E, F).$$

Any subset of $\{X, Y, Z\}$ and of $\{A, B, C\}$ can be obtained by marginalization, and the factorization

is preserved, e.g.

$$\begin{aligned}
 p(X, A, B \mid D, E, F) &= \sum_{Y, Z, C} p(X, Y, Z, A, B, C \mid D, E, F) \\
 &= \sum_{Y, Z} p(X, Y, Z \mid D, E, F) \sum_C p(A, B, C \mid D, E, F) \\
 &= p(X \mid D, E, F) p(A, B \mid D, E, F) \\
 &\implies X \perp \{A, B\} \mid \{D, E, F\}.
 \end{aligned}$$

(Replace the sums by integrals for continuous variables.) The same argument applies to every pair of subsets; this is the Decomposition property applied on both sides.

Question 12 (Lecture 5)

Figure 12.1 shows a directed graphical model with five random variables X_1, X_2, X_3, X_4, X_5 , where $X_i \in \{0, 1\}$. The respective conditional probabilities are shown in Table 12.1, where a, b, c and d are unknown values. Given that the minimal probability 0.00216 occurs at the configuration of $X_1 = 1, X_2 = 0, X_3 = 1, X_4 = 1, X_5 = 0$, and the maximal probability 0.10976 occurs at the configuration of $X_1 = 0, X_2 = 0, X_3 = 0, X_4 = 1, X_5 = 1$, find the **numerical values** of the probability distribution $p(X_2, X_3, X_5)$. **Show all your workings clearly.**

X_2	X_3	X_5	$p(X_5 X_2, X_3)$
0	0	0	0.3
0	0	1	0.7
0	1	0	0.1
0	1	1	0.9
1	0	0	0.5
1	0	1	0.5
1	1	0	0.8
1	1	1	0.2

X_2	X_3	$p(X_3 X_2)$
0	0	0.7
0	1	0.3
1	0	0.6
1	1	0.4

X_1	X_2	$p(X_2 X_1)$
0	0	0.8
0	1	0.2
1	0	c
1	1	d

X_4	X_2	X_3	X_5	$p(X_4 X_2, X_3, X_5)$
0	0	0	0	0.2
0	0	0	1	0.3
0	0	1	0	0.6
0	0	1	1	0.9
0	1	0	0	0.2
0	1	0	1	0.3
0	1	1	0	0.2
0	1	1	1	0.6
1	0	0	0	0.8
1	0	0	1	0.7
1	0	1	0	0.4
1	0	1	1	0.1
1	1	0	0	0.8
1	1	0	1	0.7
1	1	1	0	0.8
1	1	1	1	0.4

X_1	$p(X_1)$
0	a
1	b

Table 12.1

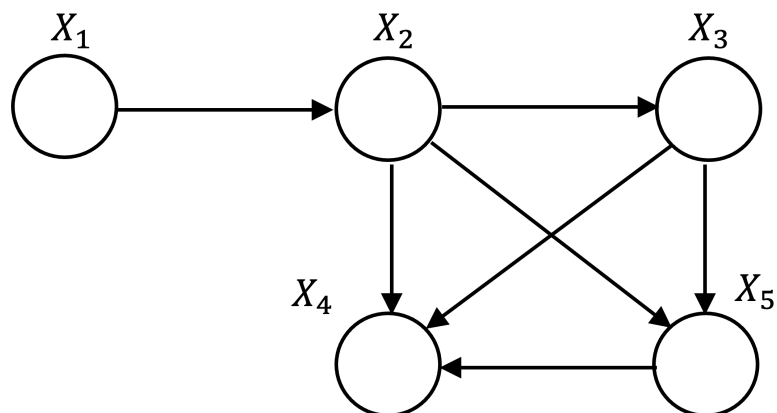


Figure 12.1

Answer:

Joint probability:

$$p(x) = p(x_1) p(x_2 | x_1) p(x_3 | x_2) p(x_4 | x_2, x_3, x_5) p(x_5 | x_2, x_3).$$

Given the minimal probability:

$$p(x_1=1) p(x_2=0 | x_1=1) p(x_3=1 | x_2=0) p(x_4=1 | x_2=0, x_3=1, x_5=0) p(x_5=0 | x_2=0, x_3=1) = 0.00216,$$

we have

$$(b)(c)(0.3)(0.4)(0.1) = 0.00216 \implies bc = 0.18. \quad (1)$$

Given the maximal probability:

$$p(x_1=0) p(x_2=0 | x_1=0) p(x_3=0 | x_2=0) p(x_4=1 | x_2=0, x_3=0, x_5=1) p(x_5=1 | x_2=0, x_3=0) = 0.10976,$$

we have

$$(a)(0.8)(0.7)(0.7)(0.7) = 0.10976 \implies a = 0.4. \quad (2)$$

Using $a + b = 1$, we get $b = 1 - 0.4 = 0.6$. Substituting b into (1): $c = 0.18/0.6 = 0.3$, and hence $d = 1 - c = 0.7$.

Solving for $p(X_2, X_3, X_5)$:

$$\begin{aligned} p(x_2, x_3, x_5) &= \sum_{x_1} \sum_{x_4} p(x_1) p(x_2 | x_1) p(x_3 | x_2) p(x_4 | x_2, x_3, x_5) p(x_5 | x_2, x_3) \\ &= p(x_3 | x_2) p(x_5 | x_2, x_3) \sum_{x_1} p(x_1) p(x_2 | x_1) \underbrace{\sum_{x_4} p(x_4 | x_2, x_3, x_5)}_{=1} \\ &= p(x_3 | x_2) p(x_5 | x_2, x_3) [p(x_1=0) p(x_2 | x_1=0) + p(x_1=1) p(x_2 | x_1=1)], \end{aligned}$$

where the bracket is $p(x_2)$: $p(x_2=0) = (0.4)(0.8) + (0.6)(0.3) = 0.5$ and $p(x_2=1) = (0.4)(0.2) + (0.6)(0.7) = 0.5$.

X_2	X_3	X_5	$p(X_2, X_3, X_5)$
0	0	0	$(0.7)(0.3)[(0.4)(0.8) + (0.6)(0.3)] = \mathbf{0.105}$
0	0	1	$(0.7)(0.7)[(0.4)(0.8) + (0.6)(0.3)] = \mathbf{0.245}$
0	1	0	$(0.3)(0.1)[(0.4)(0.8) + (0.6)(0.3)] = \mathbf{0.015}$
0	1	1	$(0.3)(0.9)[(0.4)(0.8) + (0.6)(0.3)] = \mathbf{0.135}$
1	0	0	$(0.6)(0.5)[(0.4)(0.2) + (0.6)(0.7)] = \mathbf{0.15}$
1	0	1	$(0.6)(0.5)[(0.4)(0.2) + (0.6)(0.7)] = \mathbf{0.15}$
1	1	0	$(0.4)(0.8)[(0.4)(0.2) + (0.6)(0.7)] = \mathbf{0.16}$
1	1	1	$(0.4)(0.2)[(0.4)(0.2) + (0.6)(0.7)] = \mathbf{0.04}$

(Check: the eight entries sum to 1.)

Question 13 (Lecture 3)

Prove the following conditional independences are true:

- a) $(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z)$. This is also known as **Decomposition**.
- b) $(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z, W)$. This is also known as **Weak Union**.
- c) $(X \perp Y \mid Z)$ and $(X \perp W \mid Z, Y) \implies (X \perp W, Y \mid Z)$. This is also known as **Contraction**.

Answer:

a)

$$\begin{aligned}
 p(X, Y \mid Z) &= \sum_W p(X, Y, W \mid Z) \\
 &= \sum_W p(X \mid Z) p(Y, W \mid Z) && \text{since } (X \perp Y, W \mid Z) \\
 &= p(X \mid Z) \sum_W p(Y, W \mid Z) \\
 &= p(X \mid Z) p(Y \mid Z) \implies (X \perp Y \mid Z).
 \end{aligned}$$

i.e. $(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z)$.

b)

$$\begin{aligned}
 p(X, Y, W \mid Z) &= p(X \mid Z) p(Y, W \mid Z) && \text{since } (X \perp Y, W \mid Z) \\
 &= p(X \mid Z) p(Y \mid Z, W) p(W \mid Z) && \text{(chain rule)} \\
 &= p(X \mid Z, W) p(Y \mid Z, W) p(W \mid Z).
 \end{aligned}$$

We can write $p(X \mid Z) = p(X \mid Z, W)$ because, by Decomposition (part a, with the roles of Y and W swapped), $(X \perp Y, W \mid Z) \implies (X \perp W \mid Z)$. Dividing both sides by $p(W \mid Z)$ gives $p(X, Y \mid Z, W) = p(X \mid Z, W) p(Y \mid Z, W)$, i.e. $(X \perp Y \mid Z, W)$.

Therefore, $(X \perp Y, W \mid Z) \implies (X \perp Y \mid Z, W)$.

c)

$$\begin{aligned}
 p(X, Y, W \mid Z) &= p(X \mid W, Y, Z) p(W, Y \mid Z) && \text{(chain rule)} \\
 &= p(X \mid Y, Z) p(W, Y \mid Z) && \text{since } (X \perp W \mid Z, Y) \\
 &= p(X \mid Z) p(W, Y \mid Z) && \text{since } (X \perp Y \mid Z).
 \end{aligned}$$

i.e. $(X \perp Y \mid Z)$ and $(X \perp W \mid Z, Y) \implies (X \perp W, Y \mid Z)$.

Question 14 (Lecture 6)

Figure 14.1 shows a three-node Bayesian network with random variables $X_i \in \mathbb{R}$. Furthermore,

$$p(X_1 | \mu_1, \sigma_1^2) = \mathcal{N}(X_1 | \mu_1, \sigma_1^2), \quad p(X_2 | \mu_2, \sigma_2^2) = \mathcal{N}(X_2 | \mu_2, \sigma_2^2),$$

$$p(X_3 | X_1, X_2, w_0, w_1, w_2, \sigma_3^2) = \mathcal{N}(X_3 | w_0 + w_1 X_1 + w_2 X_2, \sigma_3^2),$$

where

$$\mathcal{N}(X | \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -0.5 \frac{(X - \mu)^2}{\sigma^2} \right\}$$

is the Gaussian distribution parameterized by the mean μ and variance σ^2 . We have the linear-Gaussian distribution when the mean μ is parameterized by $w_0, \dots, w_M$ as a weighted sum of the parent nodes $\{X_{\pi,1} \dots X_{\pi,M}\}$ of X , i.e.,

$$\mu = w_0 + \sum_m w_m X_{\pi,m}.$$

Given 10 observations of the three-node Bayesian network in Table 14.1, find the probability density at $p(X_1 = 1.5, X_2 = 0.5, X_3 = 1.0)$. **Show all your workings clearly.**

Useful equation:

$$\frac{d}{dx} \ln(f(x)) = \frac{1}{f(x)} \frac{df(x)}{dx}.$$

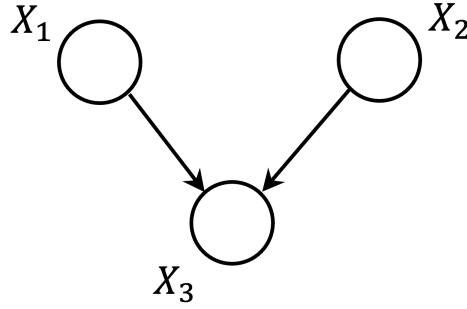


Figure 14.1

Observation #	X_1	X_2	X_3
1	1.86	-0.03	-0.31
2	1.68	1.09	1.48
3	2.41	-0.45	2.18
4	2.40	-0.33	2.98
5	0.87	0.99	3.08
6	1.96	2.83	2.24
7	1.78	0.07	-0.89
8	2.81	1.44	1.18
9	3.42	0.72	0.88
10	3.44	2.34	6.15

Table 14.1

Answer:

Joint probability:

$$p(X_1, X_2, X_3) = p(X_1 | \mu_1, \sigma_1^2) p(X_2 | \mu_2, \sigma_2^2) p(X_3 | X_1, X_2, w_0, w_1, w_2, \sigma_3^2),$$

$$\ln p(X_1, X_2, X_3) = \ln p(X_1 | \mu_1, \sigma_1^2) + \ln p(X_2 | \mu_2, \sigma_2^2) + \ln p(X_3 | X_1, X_2, w_0, w_1, w_2, \sigma_3^2).$$

The log-likelihood of the $N = 10$ i.i.d. observations decomposes into three independent terms, so each set of parameters can be estimated separately by maximum likelihood.

Parameters of X_1 :

$$\begin{aligned} \operatorname{argmax}_{\mu_1, \sigma_1^2} \sum_n \ln p(X_{1,n} | \mu_1, \sigma_1^2) &= \operatorname{argmax}_{\mu_1, \sigma_1^2} \sum_n \ln \left[\frac{1}{\sqrt{2\pi\sigma_1^2}} \exp \left\{ -0.5 \frac{(X_{1,n} - \mu_1)^2}{\sigma_1^2} \right\} \right] \\ &= \operatorname{argmax}_{\mu_1, \sigma_1^2} \underbrace{\sum_n \left\{ -0.5 \frac{(X_{1,n} - \mu_1)^2}{\sigma_1^2} - 0.5 \ln 2\pi - 0.5 \ln \sigma_1^2 \right\}}_{\mathcal{L}}. \end{aligned}$$

$$\frac{\partial \mathcal{L}}{\partial \mu_1} = \frac{\sum_n (X_{1,n} - \mu_1)}{\sigma_1^2} = 0 \implies \mu_1 = \frac{1}{N} \sum_n X_{1,n},$$

$$\frac{\partial \mathcal{L}}{\partial \sigma_1^2} = \sum_n \left\{ 0.5 \frac{(X_{1,n} - \mu_1)^2}{\sigma_1^4} - 0.5 \frac{1}{\sigma_1^2} \right\} = 0 \implies \sigma_1^2 = \frac{1}{N} \sum_n (X_{1,n} - \mu_1)^2.$$

$$\mu_1 = \frac{1.86 + 1.68 + 2.41 + 2.40 + 0.87 + 1.96 + 1.78 + 2.81 + 3.42 + 3.44}{10} = \frac{22.63}{10} = \mathbf{2.263},$$

$$\sigma_1^2 = \frac{(1.86 - 2.263)^2 + (1.68 - 2.263)^2 + \dots + (3.44 - 2.263)^2}{10} = \mathbf{0.5831}.$$

Parameters of X_2 (identical derivation):

$$\mu_2 = \frac{1}{N} \sum_n X_{2,n} = \frac{8.67}{10} = \mathbf{0.8670}, \quad \sigma_2^2 = \frac{1}{N} \sum_n (X_{2,n} - \mu_2)^2 = \mathbf{1.1045}.$$

Parameters of X_3 :

$$\begin{aligned} \operatorname{argmax}_{w_0, w_1, w_2, \sigma_3^2} \sum_n \ln p(X_{3,n} | X_{1,n}, X_{2,n}, w_0, w_1, w_2, \sigma_3^2), \\ \mathcal{L} = \sum_n \left\{ -0.5 \frac{(X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n})^2}{\sigma_3^2} - 0.5 \ln 2\pi - 0.5 \ln \sigma_3^2 \right\}. \end{aligned}$$

Setting the derivatives to zero (the common factor $1/\sigma_3^2$ cancels):

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial w_0} &\propto \sum_n (X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n}) = 0, \\ \frac{\partial \mathcal{L}}{\partial w_1} &\propto \sum_n (X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n}) X_{1,n} = 0, \\ \frac{\partial \mathcal{L}}{\partial w_2} &\propto \sum_n (X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n}) X_{2,n} = 0,\end{aligned}$$

which gives the linear system

$$\begin{bmatrix} N & \sum_n X_{1,n} & \sum_n X_{2,n} \\ \sum_n X_{1,n} & \sum_n X_{1,n} X_{1,n} & \sum_n X_{2,n} X_{1,n} \\ \sum_n X_{2,n} & \sum_n X_{1,n} X_{2,n} & \sum_n X_{2,n} X_{2,n} \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} \sum_n X_{3,n} \\ \sum_n X_{3,n} X_{1,n} \\ \sum_n X_{3,n} X_{2,n} \end{bmatrix},$$

$$\begin{bmatrix} 10 & 22.63 & 8.67 \\ 22.63 & 57.0431 & 20.99 \\ 8.67 & 20.99 & 18.5619 \end{bmatrix} \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} 18.97 \\ 47.2828 \\ 25.708 \end{bmatrix} \implies \begin{bmatrix} w_0 \\ w_1 \\ w_2 \end{bmatrix} = \begin{bmatrix} -0.0502 \\ 0.5661 \\ 0.7683 \end{bmatrix}.$$

$$\begin{aligned}\frac{\partial \mathcal{L}}{\partial \sigma_3^2} &= \sum_n \left\{ 0.5 \frac{(X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n})^2}{\sigma_3^4} - 0.5 \frac{1}{\sigma_3^2} \right\} = 0 \\ \implies \sigma_3^2 &= \frac{1}{N} \sum_n (X_{3,n} - w_0 - w_1 X_{1,n} - w_2 X_{2,n})^2 = 2.5639.\end{aligned}$$

Probability density:

$$\begin{aligned}p(X_1 = 1.5, X_2 = 0.5, X_3 = 1.0) \\ &= p(X_1 | \mu_1, \sigma_1^2) p(X_2 | \mu_2, \sigma_2^2) p(X_3 | X_1, X_2, w_0, w_1, w_2, \sigma_3^2) \\ &= \mathcal{N}(1.5 | 2.263, 0.5831) \times \mathcal{N}(0.5 | 0.867, 1.1045) \times \mathcal{N}(1.0 | \mu_3, 2.5639),\end{aligned}$$

where $\mu_3 = -0.0502 + (0.5661)(1.5) + (0.7683)(0.5) = 1.1831$. Hence

$$\begin{aligned}p(X_1 = 1.5, X_2 = 0.5, X_3 = 1.0) &= \frac{1}{\sqrt{2\pi(0.5831)}} \exp \left\{ -0.5 \frac{(1.5 - 2.263)^2}{0.5831} \right\} \\ &\quad \times \frac{1}{\sqrt{2\pi(1.1045)}} \exp \left\{ -0.5 \frac{(0.5 - 0.867)^2}{1.1045} \right\} \\ &\quad \times \frac{1}{\sqrt{2\pi(2.5639)}} \exp \left\{ -0.5 \frac{(1.0 - 1.1831)^2}{2.5639} \right\} \\ &= 0.3171 \times 0.3571 \times 0.2475 = \mathbf{0.0280}.\end{aligned}$$

Question 15 (Lecture 5 and 8)

Figure 15.1 shows a hidden Markov model over two timesteps. $X_n \in \{0, 1\}$ is a set of binary state latent random variables, and $Z_n \in \mathbb{R}$ and $Y_n \in \mathbb{R}$ are sets of continuous state random variables, where $n = 1, 2$.

- a) Find $p(X_1 \mid Y_1 = 0, Y_2 = 0, Z_1 = 1, Z_2 = 1)$ and $p(X_1, X_2 \mid Y_1 = 0, Y_2 = 0, Z_1 = 1, Z_2 = 1)$.
- b) Find the configuration of the latent random variables that gives the **minimum probability** of $p(X_1, X_2, Y_1 = 2, Y_2 = 1, Z_1 = 2, Z_2 = 0)$.

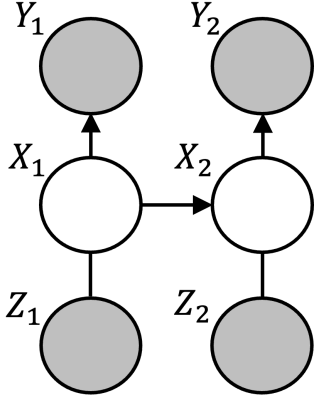


Figure 15.1

Emission Probability:

$$p(Y_n \mid X_n = 0) = \mathcal{N}(Y_n \mid \mu_0 = 0.1, \sigma_0^2 = 0.1),$$

$$p(Y_n \mid X_n = 1) = \mathcal{N}(Y_n \mid \mu_1 = 0.2, \sigma_1^2 = 0.2),$$

$$\text{where } \mathcal{N}(Y \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -0.5 \left(\frac{Y - \mu}{\sigma} \right)^2 \right\}.$$

Prior Probability:

$$p(X_1 = 0) = 0.2, \quad p(X_1 = 1) = 0.8.$$

Transition Probability:

$$p(X_n \mid X_{n-1}) = \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}.$$

Pairwise Potential:

$$\psi(Z_n, X_n = 0) = Z_n + 2, \quad \psi(Z_n, X_n = 1) = Z_n + 3.$$

Answer:

Joint probability:

$$p(X, Y, Z) = \frac{1}{Z} p(X_1) p(X_2 \mid X_1) p(Y_1 \mid X_1) p(Y_2 \mid X_2) \psi(Z_1, X_1) \psi(Z_2, X_2),$$

where the rows of the transition matrix are indexed by X_{n-1} and the columns by X_n , i.e. $p(X_2=0 \mid X_1=0) = 0.8$, $p(X_2=1 \mid X_1=0) = 0.2$, $p(X_2=0 \mid X_1=1) = 0.1$, $p(X_2=1 \mid X_1=1) = 0.9$. The normalization constant Z does not depend on X_1, X_2 , so it cancels in conditional probabilities and does not affect the argmin.

a) (i)

$$\begin{aligned} p(X_1 \mid Y_1=0, Y_2=0, Z_1=1, Z_2=1) &= \frac{p(X_1, Y_1=0, Y_2=0, Z_1=1, Z_2=1)}{p(Y_1=0, Y_2=0, Z_1=1, Z_2=1)} \\ &= \frac{p(X_1) p(Y_1=0 \mid X_1) \psi(Z_1=1, X_1) \sum_{X_2} p(X_2 \mid X_1) p(Y_2=0 \mid X_2) \psi(Z_2=1, X_2)}{\sum_{X_1} \sum_{X_2} p(X_1) p(X_2 \mid X_1) p(Y_1=0 \mid X_1) p(Y_2=0 \mid X_2) \psi(Z_1=1, X_1) \psi(Z_2=1, X_2)}, \end{aligned}$$

where the $\frac{1}{Z}$ factors in the numerator and denominator cancel. The required quantities are

$$p(Y=0 | X=0) = \frac{1}{\sqrt{2\pi(0.1)}} \exp \left\{ -0.5 \frac{(0-0.1)^2}{0.1} \right\} = 1.2000,$$

$$p(Y=0 | X=1) = \frac{1}{\sqrt{2\pi(0.2)}} \exp \left\{ -0.5 \frac{(0-0.2)^2}{0.2} \right\} = 0.8072,$$

$$\psi(Z=1, X=0) = 3, \quad \psi(Z=1, X=1) = 4.$$

Numerator. When $X_1 = 0$:

$$(0.2)(1.2000)(3) [(0.8)(1.2000)(3) + (0.2)(0.8072)(4)] = 0.7200 \times 3.5258 = 2.5387.$$

When $X_1 = 1$:

$$(0.8)(0.8072)(4) [(0.1)(1.2000)(3) + (0.9)(0.8072)(4)] = 2.5830 \times 3.2658 = 8.4355.$$

Denominator: $2.5387 + 8.4355 = 10.9742$. Therefore

$$p(X_1=0 | Y_1=0, Y_2=0, Z_1=1, Z_2=1) = \frac{2.5387}{10.9742} = \mathbf{0.2313}, \quad p(X_1=1 | \dots) = \frac{8.4355}{10.9742} = \mathbf{0.7687}.$$

a) (ii)

$$\begin{aligned} p(X_1, X_2 | Y_1=0, Y_2=0, Z_1=1, Z_2=1) \\ &= \frac{p(X_1) p(X_2 | X_1) p(Y_1=0 | X_1) p(Y_2=0 | X_2) \psi(Z_1=1, X_1) \psi(Z_2=1, X_2)}{\sum_{X_1} \sum_{X_2} (\dots)} \\ &= \frac{p(X_1) p(X_2 | X_1) p(Y_1=0 | X_1) p(Y_2=0 | X_2) \psi(Z_1=1, X_1) \psi(Z_2=1, X_2)}{10.9742}. \end{aligned}$$

X_1	X_2	Numerator	$p(X_1, X_2 Y_1=0, Y_2=0, Z_1=1, Z_2=1)$
0	0	$(0.2)(0.8)(1.2000)(1.2000)(3)(3) = 2.0737$	$2.0737/10.9742 = \mathbf{0.1890}$
0	1	$(0.2)(0.2)(1.2000)(0.8072)(3)(4) = 0.4649$	$0.4649/10.9742 = \mathbf{0.0424}$
1	0	$(0.8)(0.1)(0.8072)(1.2000)(4)(3) = 0.9299$	$0.9299/10.9742 = \mathbf{0.0847}$
1	1	$(0.8)(0.9)(0.8072)(0.8072)(4)(4) = 7.5056$	$7.5056/10.9742 = \mathbf{0.6839}$

b)

$$\operatorname{argmin}_{X_1, X_2} p(X_1, X_2, Y_1=2, Y_2=1, Z_1=2, Z_2=0).$$

The required quantities are

$$p(Y=2 | X=0) = \frac{1}{\sqrt{2\pi(0.1)}} \exp \left\{ -0.5 \frac{(2-0.1)^2}{0.1} \right\} = 1.8277 \times 10^{-8},$$

$$p(Y=2 | X=1) = \frac{1}{\sqrt{2\pi(0.2)}} \exp \left\{ -0.5 \frac{(2-0.2)^2}{0.2} \right\} = 2.7078 \times 10^{-4},$$

$$p(Y=1 | X=0) = \frac{1}{\sqrt{2\pi(0.1)}} \exp \left\{ -0.5 \frac{(1-0.1)^2}{0.1} \right\} = 0.02198,$$

$$p(Y=1 | X=1) = \frac{1}{\sqrt{2\pi(0.2)}} \exp \left\{ -0.5 \frac{(1-0.2)^2}{0.2} \right\} = 0.18010,$$

$$\psi(Z=2, X=0) = 4, \quad \psi(Z=2, X=1) = 5, \quad \psi(Z=0, X=0) = 2, \quad \psi(Z=0, X=1) = 3.$$

Hence (up to the constant $1/Z$):

X_1	X_2	$p(X_1) p(X_2 X_1) p(Y_1=2 X_1) p(Y_2=1 X_2) \psi(Z_1=2, X_1) \psi(Z_2=0, X_2)$
0	0	$(0.2)(0.8)(1.8277 \times 10^{-8})(0.02198)(4)(2) = \mathbf{5.1419 \times 10^{-10}}$
0	1	$(0.2)(0.2)(1.8277 \times 10^{-8})(0.18010)(4)(3) = 1.5800 \times 10^{-9}$
1	0	$(0.8)(0.1)(2.7078 \times 10^{-4})(0.02198)(5)(2) = 4.7612 \times 10^{-6}$
1	1	$(0.8)(0.9)(2.7078 \times 10^{-4})(0.18010)(5)(3) = 5.2669 \times 10^{-4}$

Minimum probability: $\mathbf{X_1 = 0, X_2 = 0}$.

Question 16 (Lecture 3, 4 and 5)

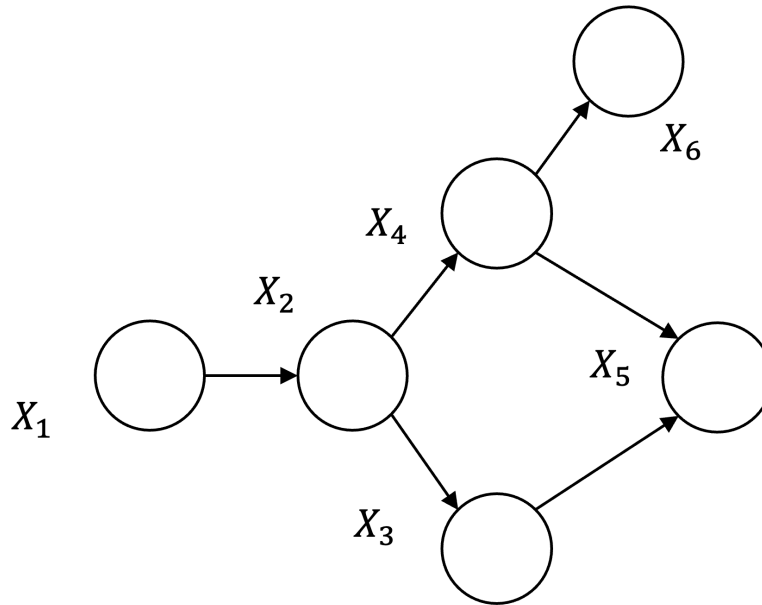


Figure 16.1

a) Prove the following conditional independences are true:

$$(i) (A \perp B, C \mid D) \implies (A \perp B \mid D),$$

$$(ii) (A \perp B, C \mid D) \implies (A \perp B \mid D, C),$$

$$(iii) (A \perp B \mid D) \text{ and } (A \perp C \mid B, D) \implies (A \perp B, C \mid D),$$

$$(iv) (A \perp B \mid C, D) \text{ and } (A \perp C \mid B, D) \implies (A \perp B, C \mid D).$$

b) Find the junction tree of the Bayesian network shown in Figure 16.1 using the following elimination order: $\{X_6, X_5, X_4, X_3, X_2, X_1\}$.

Answer:

a) (i) (Decomposition)

$$\begin{aligned} p(A, B \mid D) &= \sum_C p(A, B, C \mid D) \\ &= \sum_C p(A \mid D) p(B, C \mid D) && \text{since } (A \perp B, C \mid D) \\ &= p(A \mid D) \sum_C p(B, C \mid D) = p(A \mid D) p(B \mid D) \implies (A \perp B \mid D). \end{aligned}$$

a) (ii) (Weak Union) By the same argument as in (i), summing over B instead of C :

$$p(A, C \mid D) = \sum_B p(A \mid D) p(B, C \mid D) = p(A \mid D) p(C \mid D) \implies (A \perp C \mid D),$$

so $p(A \mid D) = p(A \mid D, C)$. Then

$$\begin{aligned} p(A, B, C \mid D) &= p(A \mid D) p(B, C \mid D) && \text{since } (A \perp B, C \mid D) \\ &= p(A \mid D) p(B \mid D, C) p(C \mid D) && \text{(chain rule)} \\ &= p(A \mid D, C) p(B \mid D, C) p(C \mid D) && \text{since } (A \perp C \mid D). \end{aligned}$$

Dividing by $p(C \mid D)$: $p(A, B \mid D, C) = p(A \mid D, C) p(B \mid D, C)$, i.e. $(A \perp B, C \mid D) \implies (A \perp B \mid D, C)$.

a) (iii) (Contraction)

$$\begin{aligned} p(A, B, C \mid D) &= p(A \mid B, C, D) p(B, C \mid D) && \text{(chain rule)} \\ &= p(A \mid B, D) p(B, C \mid D) && \text{since } (A \perp C \mid B, D) \\ &= p(A \mid D) p(B, C \mid D) && \text{since } (A \perp B \mid D), \end{aligned}$$

i.e. $(A \perp B \mid D) \text{ and } (A \perp C \mid B, D) \implies (A \perp B, C \mid D)$.

a) (iv) (Intersection; this holds for strictly positive distributions $p > 0$.) We have

$$\begin{aligned} p(A \mid B, C, D) &= p(A \mid C, D) && \text{since } A \perp B \mid C, D, \\ p(A \mid B, C, D) &= p(A \mid B, D) && \text{since } A \perp C \mid B, D. \end{aligned}$$

Hence $p(A \mid C, D) = p(A \mid B, D)$ for all values of B and C (all such conditionals are well defined because

$p > 0$). The left-hand side does not depend on B , and the right-hand side does not depend on C , so both depend only on (A, D) . There is therefore some function $h(A, D)$ such that

$$p(A \mid B, C, D) = p(A \mid C, D) = p(A \mid B, D) = h(A, D).$$

Moreover,

$$p(A \mid D) = \sum_{B, C} p(A \mid B, C, D) p(B, C \mid D) = h(A, D) \sum_{B, C} p(B, C \mid D) = h(A, D).$$

Thus $p(A \mid B, C, D) = h(A, D) = p(A \mid D)$, which implies $(A \perp B \mid C, D)$ and $(A \perp C \mid B, D) \implies (A \perp B, C \mid D)$.

Remark: positivity is needed. For example, if $A = B = C$ (one fair coin copied three times) and $D = \emptyset$, both premises hold, but $A \not\perp \{B, C\}$.

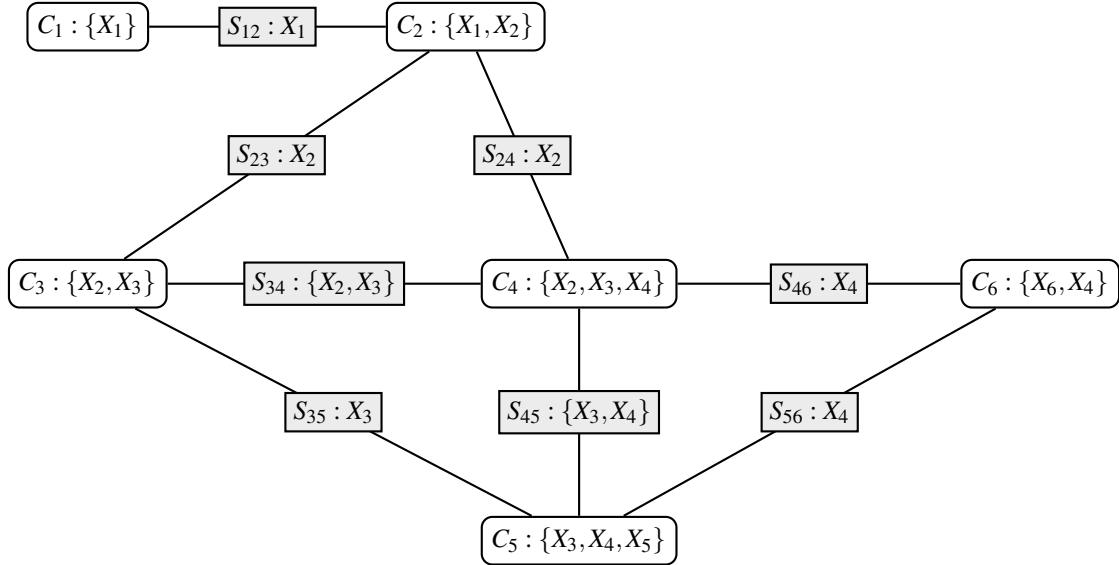
b) Moralization: drop all edge directions and marry all parents. X_5 has parents $\{X_3, X_4\}$, so the edge $X_3 - X_4$ is added. The moral graph has the edges $X_1 - X_2, X_2 - X_3, X_2 - X_4, X_3 - X_4, X_3 - X_5, X_4 - X_5, X_4 - X_6$.

Triangulation with the elimination order $\{X_6, X_5, X_4, X_3, X_2, X_1\}$: eliminating X_6 (neighbour X_4), X_5 (neighbours $\{X_3, X_4\}$, already connected), X_4 (neighbours $\{X_2, X_3\}$, already connected), X_3, X_2, X_1 adds no fill-in edges, since the moral graph is already triangulated.

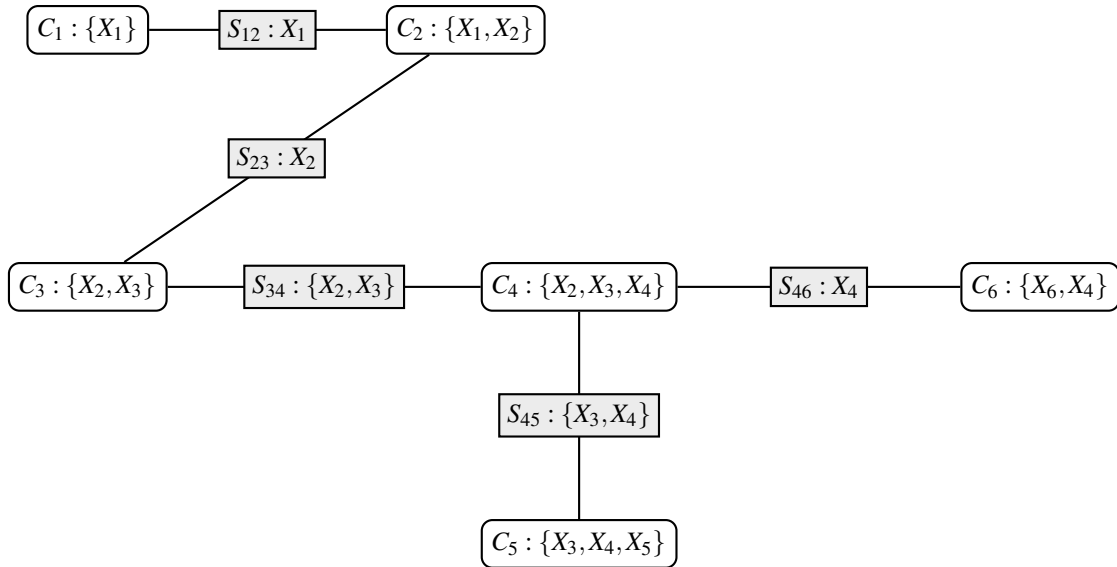
Get all clusters and all possible sepsets:

$$C_6 : \{X_6, X_4\}, \quad C_5 : \{X_3, X_4, X_5\}, \quad C_4 : \{X_2, X_3, X_4\}, \quad C_3 : \{X_2, X_3\}, \quad C_2 : \{X_1, X_2\}, \quad C_1 : \{X_1\}.$$

Fully connected cluster graph (edges with non-empty sepsets):



Maximum spanning tree, i.e. the junction tree:



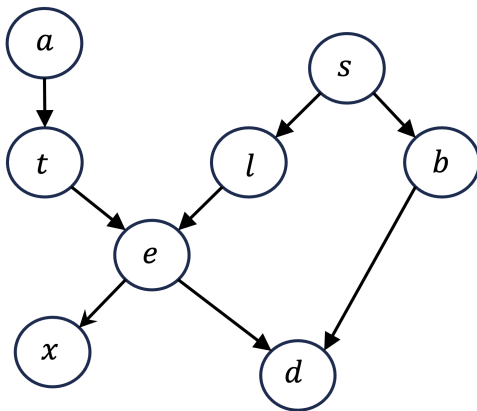
Note: C_1 and C_3 are subsets of neighbouring clusters and can be absorbed. The junction tree over the maximal cliques is then $\{X_1, X_2\} \xrightarrow{X_2} \{X_2, X_3, X_4\}$, with $\{X_3, X_4, X_5\}$ (sepset $\{X_3, X_4\}$) and $\{X_4, X_6\}$ (sepset X_4) attached to $\{X_2, X_3, X_4\}$.

Question 17 (Lectures 2 and 3)

a) **Prove** the following statements to be either true or false:

- (i) $X \perp \{A, B\} \implies X \perp A$ and $X \perp B$
- (ii) $X \perp \{A, B\} \iff X \perp A \mid B$ and $X \perp B \mid A$
- (iii) $X \perp A \mid B$ and $X \perp B \iff X \perp \{A, B\}$
- (iv) $X \perp Y \mid \{W, Z\}$ and $X \perp W \mid \{Y, Z\} \iff X \perp \{Y, W\} \mid Z$

b) Figure 17.1 shows the Chest Clinic network which concerns the diagnosis of lung disease (Tuberculosis, lung cancer, or both, or neither). In this model, a visit to Asia is assumed to increase the probability of tuberculosis.



x :	Positive X-ray
d :	Dyspnea (Shortness of Breath)
e :	Either tuberculosis or lung cancer
t :	Tuberculosis
l :	Lung Cancer
b :	Bronchitis
a :	Visited Asia
s :	Smoker

Figure 17.1: Belief Network for Chest Clinic

State **with reasoning** if the following conditional independence relations are true or false:

- (i) Tuberculosis $\perp$ Smoking $\mid$ Shortness of Breath
- (ii) Lung Cancer $\perp$ Bronchitis $\mid$ Smoking
- (iii) Visit to Asia $\perp$ Smoking $\mid$ Lung Cancer
- (iv) Visit to Asia $\perp$ Smoking $\mid$ Lung Cancer, Shortness of Breath

Show all your workings clearly.

Answer:

a) (i) True (Decomposition).

$$X \perp \{A, B\} \implies p(X, A, B) = p(X) p(A, B),$$

$$p(X, A) = \int p(X, A, B) dB = \int p(X) p(A, B) dB = p(X) p(A) \implies X \perp A.$$

$X \perp B$ can be shown in a similar way (integrate over A). □

The converse is false, i.e. $X \perp A$ and $X \perp B \not\Rightarrow X \perp \{A, B\}$. Counter-example (XOR): let A, B be independent fair coins and $X = A \oplus B$:

A	B	X	$p(A, B, X)$	$p(X \mid A, B)$
0	0	0	0.25	1
0	0	1	0	0
0	1	0	0	0
0	1	1	0.25	1
1	0	0	0	0
1	0	1	0.25	1
1	1	0	0.25	1
1	1	1	0	0

Here $p(X \mid A) = p(X \mid B) = p(X) = 0.5$ for all values, so $X \perp A$ and $X \perp B$. But $p(X \mid A, B) \in \{0, 1\} \neq p(X)$, so $X \not\perp \{A, B\}$. □

a) (ii) True for strictly positive distributions ($\Rightarrow$ always holds; $\Leftarrow$ is the Intersection property and requires $p > 0$).

($\Rightarrow$, Weak Union) $X \perp \{A, B\} \Rightarrow p(X) = p(X \mid A, B)$. By Decomposition we also have $X \perp A$, i.e. $p(X) = p(X \mid A)$. Combining the two gives $p(X \mid A, B) = p(X \mid A) \Rightarrow X \perp B \mid A$. $X \perp A \mid B$ is shown in the same way. □

($\Leftarrow$)

$$\left. \begin{array}{l} X \perp A \mid B \Rightarrow p(X \mid A, B) = p(X \mid B) \\ X \perp B \mid A \Rightarrow p(X \mid A, B) = p(X \mid A) \end{array} \right\} p(X \mid A, B) = p(X \mid A) = p(X \mid B).$$

If $p > 0$, this holds for every pair (a, b) . Then $p(X \mid A=a)$ does not depend on a (it equals $p(X \mid B=b)$ for

any fixed b), so

$$p(X) = \int p(X|A)p(A)dA = p(X|B) \int p(A)dA = p(X|B) \Rightarrow p(X|A,B) = p(X|A) = p(X|B) = p(X),$$

i.e. $X \perp \{A,B\}$. $\square$

Remark: without positivity, $\Leftarrow$ fails. For example, if $X = A = B$ (one fair coin copied three times), then $X \perp A|B$ and $X \perp B|A$ hold trivially, but $X \not\perp \{A,B\}$.

a) (iii) True.

($\Rightarrow$, Contraction) $X \perp A|B$ and $X \perp B \Rightarrow p(X|A,B) = p(X|B) = p(X) \Rightarrow X \perp \{A,B\}$. $\square$

($\Leftarrow$, Decomposition + Weak Union) $X \perp \{A,B\} \Rightarrow p(X|A,B) = p(X)$, and

$$p(X|B) = \frac{\int p(X,A,B)dA}{p(B)} = \frac{\int p(X|A,B)p(A,B)dA}{p(B)} = \frac{p(X) \int p(A,B)dA}{p(B)} = \frac{p(X)p(B)}{p(B)} = p(X).$$

Hence $p(X|A,B) = p(X|B) = p(X)$, i.e. $X \perp A|B$ and $X \perp B$. $\square$

a) (iv) True for strictly positive distributions ($\Leftarrow$ always holds; $\Rightarrow$ is the Intersection property and requires $p > 0$).

($\Rightarrow$)

$$\left. \begin{array}{l} X \perp Y | \{W,Z\} \Rightarrow p(X|W,Y,Z) = p(X|W,Z) \\ X \perp W | \{Y,Z\} \Rightarrow p(X|W,Y,Z) = p(X|Y,Z) \end{array} \right\} \Rightarrow p(X|W,Z) = p(X|Y,Z).$$

If $p > 0$, this holds for all (w,y,z) , so $p(X|W,Z)$ does not depend on W and

$$\begin{aligned} p(X|Z) &= \int p(X,W|Z)dW && \text{(marginalization)} \\ &= \int p(X|W,Z)p(W|Z)dW && \text{(product rule)} \\ &= p(X|Y,Z) \int p(W|Z)dW = p(X|Y,Z). \end{aligned}$$

Hence $p(X|W,Y,Z) = p(X|W,Z) = p(X|Y,Z) = p(X|Z) \Rightarrow X \perp \{Y,W\} | Z$. $\square$

($\Leftarrow$, Weak Union) $X \perp \{Y,W\} | Z \Rightarrow p(X|Y,W,Z) = p(X|Z)$, and

$$p(X|W,Z) = \frac{\int p(X,Y,W,Z)dY}{p(W,Z)} = \frac{\int p(X|Y,W,Z)p(Y,W,Z)dY}{p(W,Z)} = \frac{p(X|Z)p(W,Z)}{p(W,Z)} = p(X|Z).$$

So $p(X|Y,W,Z) = p(X|Z) = p(X|W,Z)$, i.e. $X \perp Y | \{W,Z\}$. Similarly,

$$p(X|Y,Z) = \frac{\int p(X|Y,W,Z)p(Y,W,Z)dW}{p(Y,Z)} = \frac{p(X|Z)p(Y,Z)}{p(Y,Z)} = p(X|Z),$$

so $p(X|Y,W,Z) = p(X|Y,Z)$, i.e. $X \perp W | \{Y,Z\}$. $\square$

Remark: as in (ii), $\Rightarrow$ fails without positivity (e.g. $X = Y = W$ with Z independent of them).

b) (i) False: Tuberculosis (t) $\perp$ Smoking (s) $|$ Shortness of Breath (d). The observed d is a descendant of

the head-to-head node e in $t \rightarrow e \leftarrow l$, which opens the path $t \rightarrow e \leftarrow l \leftarrow s$. (The path $t \rightarrow e \rightarrow d \leftarrow b \leftarrow s$ is also open, because the head-to-head node d is observed.)

b) (ii) True: Lung Cancer (l) $\perp$ Bronchitis (b) | Smoking (s). The observed s is a tail-to-tail node, which blocks the path $l \leftarrow s \rightarrow b$. The unobserved d is a head-to-head node, which blocks the path $l \rightarrow e \rightarrow d \leftarrow b$.

b) (iii) True: Visit to Asia (a) $\perp$ Smoking (s) | Lung Cancer (l). The path $a \rightarrow t \rightarrow e \leftarrow l \leftarrow s$ is blocked by the unobserved head-to-head node e (none of its descendants x, d is observed), and also by the observed l . The path $a \rightarrow t \rightarrow e \rightarrow d \leftarrow b \leftarrow s$ is blocked by the unobserved head-to-head node d .

b) (iv) False: Visit to Asia (a) $\perp$ Smoking (s) | Lung Cancer (l), Shortness of Breath (d). The head-to-head node d on the path $a \rightarrow t \rightarrow e \rightarrow d \leftarrow b \leftarrow s$ is now observed, and the other nodes on this path (t, e, b) are unobserved. So this path is open. (The path through $e \leftarrow l \leftarrow s$ remains blocked by the observed l .)

Question 18 (Lecture 6)

- (i) Consider a Bayesian network consisting of three binary variables arranged in a directed chain $A \rightarrow B \rightarrow C$, where $A, B, C \in \{0, 1\}$. The network is trained from $N = 6$ fully observed data samples as shown below:

Sample #	1	2	3	4	5	6
A	0	0	1	1	1	1
B	0	1	0	1	1	0
C	0	1	0	1	0	0

Let us denote: $\theta_A = p(A = 1)$, $\theta_{B|A=a} = p(B = 1 | A = a)$, $\theta_{C|B=b} = p(C = 1 | B = b)$, where $a, b \in \{0, 1\}$. The full parameter set of the network is thus:

$$\theta = \{\theta_A, \theta_{B|A=0}, \theta_{B|A=1}, \theta_{C|B=0}, \theta_{C|B=1}\}.$$

- Derive the MLE estimates of θ and compute the numerical values.
- Suppose we impose Beta priors $p(\theta | \alpha, \beta) = \text{Beta}(\alpha, \beta) \propto \theta^{\alpha-1} (1 - \theta)^{\beta-1}$, where $\alpha, \beta > 0$ are the hyperparameters given by:

$$\theta_A \sim \text{Beta}(2, 2), \quad \theta_{B|A=0}, \theta_{B|A=1} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(3, 2), \quad \theta_{C|B=0}, \theta_{C|B=1} \stackrel{\text{i.i.d.}}{\sim} \text{Beta}(1, 3).$$

Derive the MAP estimates of θ and compute the numerical values.

- Comment briefly on how the MAP estimates differ from the MLE in this example and explain why the difference arises.
- (ii) Now suppose variable C from (i) is connected to a continuous variable $Y \in \mathbb{R}$ through an undirected edge forming a hybrid network: $A \rightarrow B \rightarrow C - Y$. Conditional on C , the variable Y follows a Gaussian:

$$p(Y | C = c) = \mathcal{N}(\mu_c, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - \mu_c)^2}{2\sigma^2}\right),$$

where $\sigma^2 = 1$. We observe the following continuous values corresponding to the six samples above:

Sample #	1	2	3	4	5	6
C	0	1	0	1	0	0
Y	0.1	2.2	-0.3	2.0	0.2	0.0

- a) Derive the MLE estimates $\mu_{c=0}$ and $\mu_{c=1}$, and compute the numerical values.
- b) Using the MLE estimates from part (a), derive the predictive distribution for a new observation Y_{new} given each class label $C = c$. Express $p(Y_{\text{new}} | C = c)$ explicitly, and provide the numerical values of the predicted mean and variance for each class.

Show all your workings clearly.

Answer:

(i) a) MLE. The likelihood of the N i.i.d. samples factorizes according to the network:

$$\mathcal{L}(\theta) = \prod_{n=1}^N p(a_n) p(b_n | a_n) p(c_n | b_n) \implies \ln \mathcal{L}(\theta) = \sum_n \ln p(a_n) + \sum_n \ln p(b_n | a_n) + \sum_n \ln p(c_n | b_n).$$

Each parameter appears in only one group of terms, so each can be maximized separately. Every term is a Bernoulli log-likelihood of the form

$$\ell(\theta) = N_1 \ln \theta + N_0 \ln(1 - \theta),$$

where N_1 (N_0) is the number of relevant samples in which the child variable equals 1 (0). Setting the derivative to zero:

$$\frac{d\ell}{d\theta} = \frac{N_1}{\theta} - \frac{N_0}{1-\theta} = 0 \implies \boxed{\hat{\theta}_{\text{MLE}} = \frac{N_1}{N_1 + N_0}}$$

The counts from the data are:

Parameter	Relevant samples	N_1	N_0	$\hat{\theta}_{\text{MLE}}$
θ_A	all (1–6)	4	2	$4/6 = \mathbf{0.667}$
$\theta_{B A=0}$	$A = 0$: 1, 2	1	1	$1/2 = \mathbf{0.5}$
$\theta_{B A=1}$	$A = 1$: 3, 4, 5, 6	2	2	$2/4 = \mathbf{0.5}$
$\theta_{C B=0}$	$B = 0$: 1, 3, 6	0	3	$0/3 = \mathbf{0}$
$\theta_{C B=1}$	$B = 1$: 2, 4, 5	2	1	$2/3 = \mathbf{0.667}$

For $\theta_{C|B=0}$ we have $N_1 = 0$, so $\ell(\theta) = 3 \ln(1 - \theta)$ has no interior stationary point. It is strictly decreasing in θ , so the maximum is at the boundary $\hat{\theta} = 0$, which agrees with the formula $N_1/(N_1 + N_0)$.

(i) b) MAP. With a $\text{Beta}(\alpha, \beta)$ prior, the log-posterior of each parameter is (up to a constant)

$$\ln p(\theta | \mathcal{D}) = (N_1 + \alpha - 1) \ln \theta + (N_0 + \beta - 1) \ln(1 - \theta) + \text{const.}$$

$$\frac{d}{d\theta} \ln p(\theta | \mathcal{D}) = \frac{N_1 + \alpha - 1}{\theta} - \frac{N_0 + \beta - 1}{1 - \theta} = 0 \implies \boxed{\hat{\theta}_{\text{MAP}} = \frac{N_1 + \alpha - 1}{N_1 + N_0 + \alpha + \beta - 2}}$$

Parameter	Prior	(N_1, N_0)	$\hat{\theta}_{\text{MAP}}$	$\hat{\theta}_{\text{MLE}}$
θ_A	Beta(2, 2)	(4, 2)	$\frac{4+1}{6+2} = \frac{5}{8} = \mathbf{0.625}$	0.667
$\theta_{B A=0}$	Beta(3, 2)	(1, 1)	$\frac{1+2}{2+3} = \frac{3}{5} = \mathbf{0.6}$	0.5
$\theta_{B A=1}$	Beta(3, 2)	(2, 2)	$\frac{2+2}{4+3} = \frac{4}{7} = \mathbf{0.571}$	0.5
$\theta_{C B=0}$	Beta(1, 3)	(0, 3)	$\frac{0+0}{3+2} = \mathbf{0}$	0
$\theta_{C B=1}$	Beta(1, 3)	(2, 1)	$\frac{2+0}{3+2} = \frac{2}{5} = \mathbf{0.4}$	0.667

(The formula is valid when $N_1 + \alpha - 1 \geq 0$ and $N_0 + \beta - 1 \geq 0$. For $\theta_{C|B=0}$, $N_1 + \alpha - 1 = 0$, so the posterior is $\propto (1 - \theta)^5$. This is strictly decreasing, so the mode is at the boundary $\theta = 0$, again agreeing with the formula.)

(i) c) Comment. The MAP estimate is the mode of the posterior. The prior acts as $\alpha - 1$ pseudo-counts of “1” and $\beta - 1$ pseudo-counts of “0”, added to the observed counts. Each MAP estimate is therefore pulled from the MLE towards the prior mode $(\alpha - 1)/(\alpha + \beta - 2)$:

- θ_A : Beta(2, 2) has mode 0.5, so $0.667 \rightarrow 0.625$.
- $\theta_{B|A}$: Beta(3, 2) has mode $2/3$, so $0.5 \rightarrow 0.6$ and $0.5 \rightarrow 0.571$. The shift is larger for $A = 0$, which has only 2 samples, than for $A = 1$, which has 4.
- $\theta_{C|B}$: Beta(1, 3) has mode 0 and favours $C = 0$, so $0.667 \rightarrow 0.4$. For $\theta_{C|B=0}$ both estimates are 0: the data contain no $C = 1$ given $B = 0$, and with $\alpha = 1$ the prior adds no pseudo-count for $C = 1$. So this prior does not remove the zero-probability estimate (a prior with $\alpha > 1$ would).

The differences are large because $N = 6$ is small and each conditional parameter is estimated from only 2–4 samples, so the prior pseudo-counts are comparable to the data counts. As $N \rightarrow \infty$, the MAP estimates converge to the MLE.

(ii) a) MLE of μ_c . The log-likelihood of Y given C (with $\sigma^2 = 1$) is

$$\ln \mathcal{L}(\mu_0, \mu_1) = \sum_n \ln \mathcal{N}(y_n | \mu_{c_n}, \sigma^2) = \sum_{n: c_n=0} \left[-\frac{(y_n - \mu_0)^2}{2\sigma^2} \right] + \sum_{n: c_n=1} \left[-\frac{(y_n - \mu_1)^2}{2\sigma^2} \right] - \frac{N}{2} \ln(2\pi\sigma^2).$$

$$\frac{\partial \ln \mathcal{L}}{\partial \mu_c} = \frac{1}{\sigma^2} \sum_{n: c_n=c} (y_n - \mu_c) = 0 \implies \hat{\mu}_c = \frac{1}{N_c} \sum_{n: c_n=c} y_n$$

where N_c is the number of samples with $C = c$ ($N_0 = 4$: samples 1, 3, 5, 6; $N_1 = 2$: samples 2, 4).

$$\hat{\mu}_{c=0} = \frac{0.1 + (-0.3) + 0.2 + 0.0}{4} = \frac{0}{4} = \mathbf{0}, \quad \hat{\mu}_{c=1} = \frac{2.2 + 2.0}{2} = \mathbf{2.1}.$$

(ii) b) Predictive distribution. Plugging the MLE into the class-conditional Gaussian (with the known $\sigma^2 = 1$):

$$p(Y_{\text{new}} | C = c) = \mathcal{N}(Y_{\text{new}} | \hat{\mu}_c, \sigma^2) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(Y_{\text{new}} - \hat{\mu}_c)^2}{2}\right),$$

$$p(Y_{\text{new}} | C = 0) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{Y_{\text{new}}^2}{2}\right), \quad p(Y_{\text{new}} | C = 1) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(Y_{\text{new}} - 2.1)^2}{2}\right).$$

Class	Predicted mean	Predicted variance
$C = 0$	0	1
$C = 1$	2.1	1

A new observation near 0 is thus more likely to come from $C = 0$, and one near 2.1 from $C = 1$.

Useful Equations:

1. $\frac{\partial \ln f(x)}{\partial x} = \frac{1}{f(x)} \frac{\partial f(x)}{\partial x}.$
2. $\frac{\partial \exp f(x)}{\partial x} = \frac{\partial f(x)}{\partial x} \exp f(x).$

—End—